MTCEWE - MikroTik Certified Enterprise Wireless Engineer

Wireless routers
RouterBOARD hardware with integrated wireless
https://mikrotik.com/products/group/wireless-for-home-and-office
https://mikrotik.com/products/group/wireless-systems

MikroTik wireless cards
https://mikrotik.com/product/R52H

RF(Radio Frequency) Wireless Characteristics

The RF part of the electromagnetic spectrum is generally defined as that part of the spectrum where electromagnetic waves have frequencies in the range of about 3 kilohertz (3 kHz) to 300 gigahertz (300 GHz)
Probably the most important use for RF energy is in providing telecommunications services. Radio and television broadcasting, cellular telephones, radio communications for police and fire departments, amateur radio, microwave point-to-point links, and satellite communications are just a few of the many telecommunications applications.

Decibels

If the noise level is too high, it can result in degraded strength and performance for your wireless signal strength.
Noise level is measured in -dBm format (0 to -100). This is the power ratio in decibels (dB) of the measured power referenced to one milliwatt.
The closer the value to 0, the greater the noise level.
Negative values indicate less background noise. For example, -96dBm is a lower noise level than -20dBm.

Antenna theory and examples of use

An Antenna (or sometimes called as an Aerial), is an electrical device that converts electric power into electromagnetic waves (or simply radio waves) and vice-versa.
Antenna can be used for both Transmission and Reception of electromagnetic radiation i.e. a Transmitting Antenna with collect electrical signals from a transmission line and converts them into radio waves whereas a Receiving Antenna does the exact opposite i.e. it accepts radio waves from the space and converts them to electrical signals and gives them to a transmission line.
Why do we need Antennas?
There are several reasons as to why we need or why we use antennas, but an important reason as to why we use antennas is that they provide a simple way to transfer signals (or data) where other methods are impossible.

For example, take the case of an aeroplane. The pilot needs to frequently communicate with the ATC personnel. If would not make any sense if we tie up a cable (of dynamically variable length) to the tail of the plane and connect it to the ATC.
Isotropic an antenna which radiates energy in all directions equally i.e. an omnidirectional antenna is called as Isotropic Antenna. This is just a hypothetical situation and an isotropic antenna doesn't exist in practice.

Directional
A class of directional antennas is Aperture Antennas, Aperture Antennas have an opening in the surface. Usually, aperture antenna consists of Dipole or Loop Antenna in a guiding structure with an opening to emit radio waves.

Omnidirectional
Omni WiFi antennas, broadcast / propagate a full 360 degree radiation pattern, completely surrounding and saturating the radio signal in all directions outwards from the center of the antenna.
Omni WiFi antennas are generally used in public WiFi applications, farms and ranches, public parks, zoos and recreation centers, along with most RV Parks and Marinas.

Polarization
The simplest way to describe polarization is the direction in which the electric field of a radio wave oscillates while it propagates through a medium. The point of reference for specifying a polarization is looking at it from the transmitter of the signal. This can be visualized by imagining standing directly behind a radio antenna and looking in the direction it is aimed. In the case of a horizontal polarization, the electric field will move sideways in a horizontal plane. Conversely, for vertical polarization, the electric field will oscillate up and down in a vertical plane.

Attenuation
Attenuation is a reduction of signal strength during transmission,such as when sending data collected through automated monitoring. Attenuation is represented in decibels (dB), which is ten times the logarithm of the signal power at a particular input divided by the signal power at an output of a specified medium.For example, an office wall (the specific medium) that changes the propagation of an RF signal from a power level of 10 milliwatts (the input) to 5 milliwatts (the output) represents 3 dB of attenuation. Consequently, positive attenuation causes signals to become weaker when travelling through the medium.

2.4/5GHz indoor/outdoor cell sizes and transmitter powers
The transmitted power is Pt, and the received power is Pr
Maximum transmit power of
802.11 AP is Pt = 24 dBm

The Received Power Pr is given by: Pr =Pt – Path Loss

Client roaming

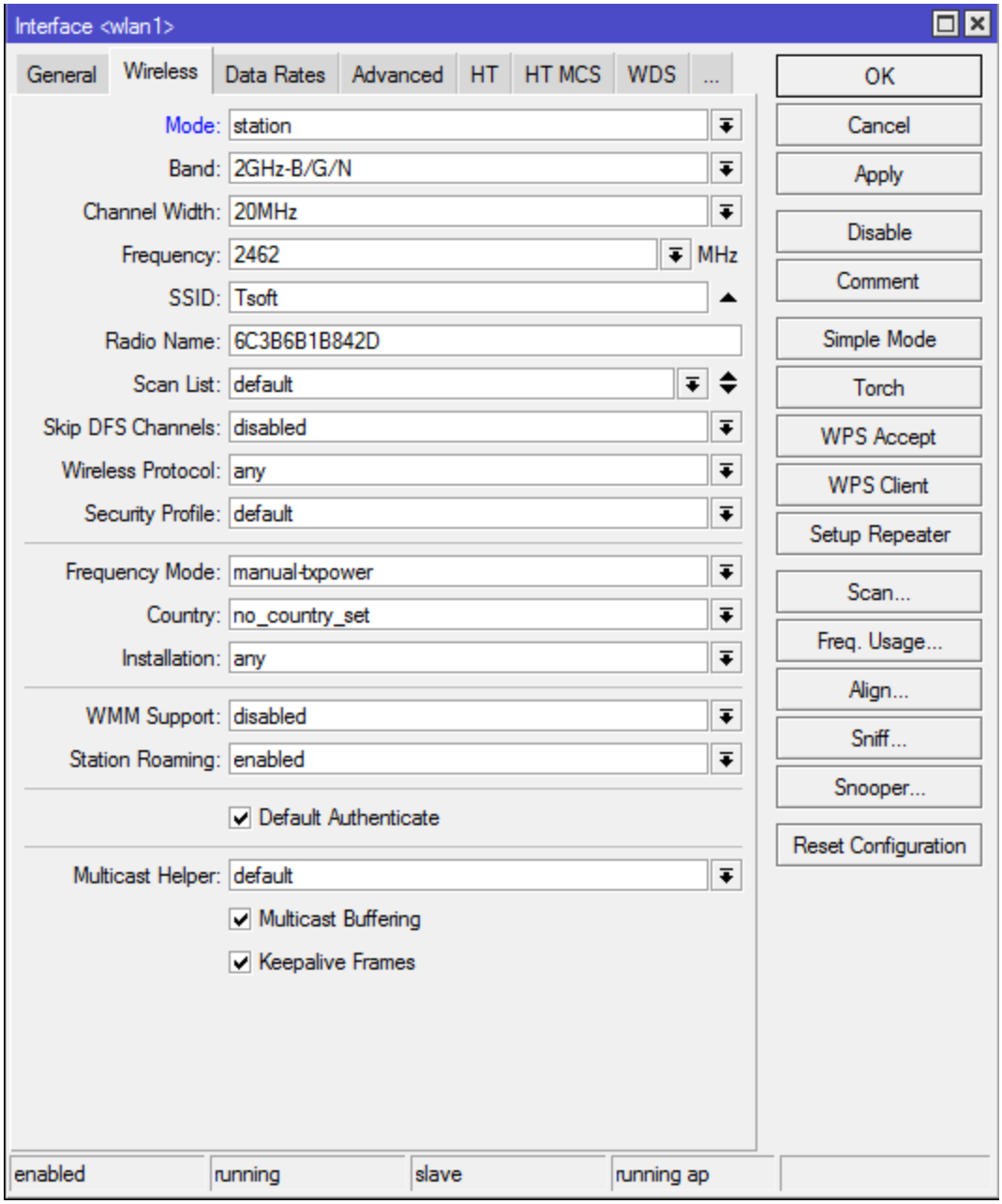
Roaming is a client side decision in 802.11 WiFi. Client devices listen for beacon frames or send probe requests to discover APs advertising the preferred SSID. The clients driver uses the received signal strength of beacons or probe responses to make decisions on whether to change APs or remain connected to the current AP. In terms of roaming, there are several points to keep in mind:

/interface wireless access-list
add authentication=no forwarding=no interface=AP signal-range=-120..-80
add interface=AP signal-range=-79..120

Station-Roaming

Station Roaming feature is available only for 802.11 wireless protocol and only for station modes. When RouterOS wireless client is connected to the AP using 802.11 wireless protocol it will periodically perform the background scan with specific time intervals. When the background scan will find an AP with better signal it will try to roam to that AP. The time intervals between the background scans will become shorter when the wireless signal becomes worse and the background scan interval will become longer when the wireless client signal will get better.

Station Roaming feature is available only for 802.11 wireless protocol and only for station modes

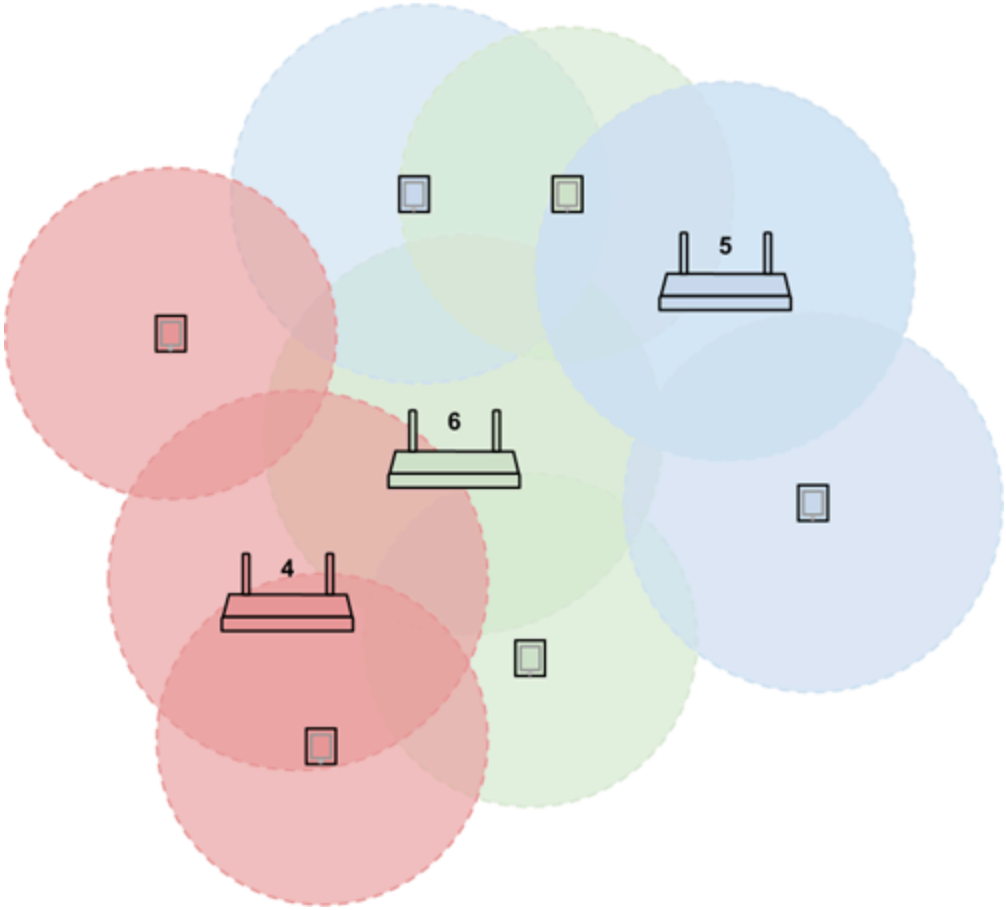


Co-channel and Adjacent-channel interference

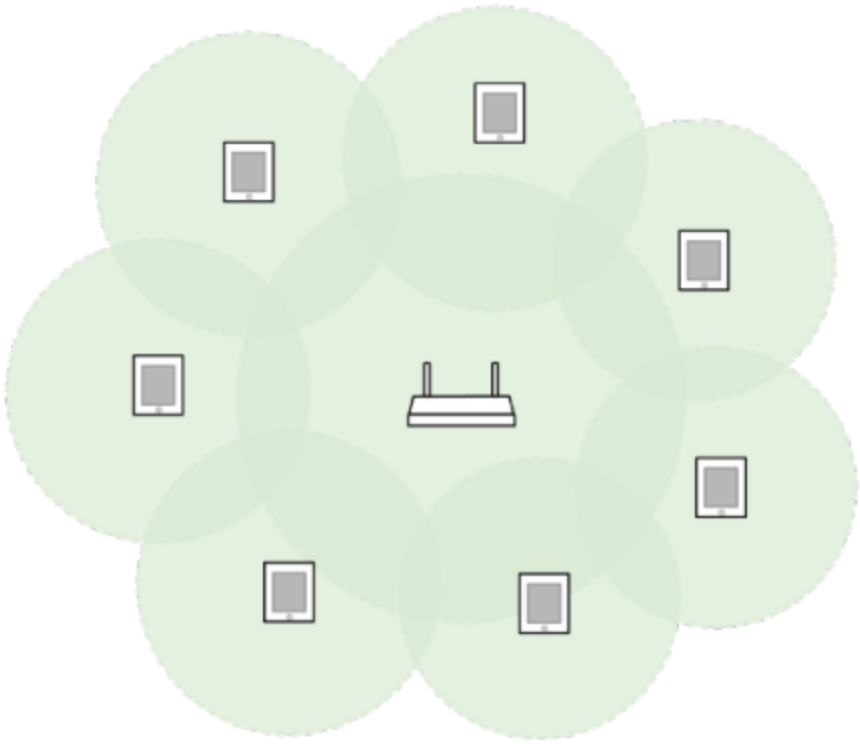
Adjacent channel congestion is the worst type of WiFi interference. To illustrate, think about being at a concert – there's a band playing really loud, and tons of people, each with their own group of friends. With this much going on, it's difficult to talk to your friends, and when you start to talk louder, the person next to you has to raise his voice to talk to their group. You're hearing multiple conversations happening, as well as music from the band, and it seems impossible to

communicate. This is exactly what happens to wireless devices trying to communicate in a congested environment.

The diagram below shows a model of how the above conversation scenario looks when access points on channel 4 (red), channel 6 (green), and channel 5 (blue) are all active at once. As one of these APs tries to talk to its clients, its transmissions become garbled because of the transmissions of the other two. This harms the performance of all of the networks.



In order to explain co-channel congestion, we'll move our imaginary conversation from a concert venue to a classroom. Think back to your school days – chances are you can think of at least one class that had a student who would talk slower than the other kids, and everyone else would have to wait for their turn to ask a question. Co-channel congestion works in a similar manner: the performance is hindered by the wait times, but the bandwidth is managed, and every device will get a chance to talk to its associated AP. The diagram below depicts a wireless access point and its associated clients, which can only talk one at a time.



Choosing correct access point placement
access points should typically be installed below a ceiling in a location that will boost signal strength in every key portion of the area.
Building materials

- Brick Wall (8 dB)
- Cinder wall (4 dB)
- Concrete Wall (12 dB)
- Dry Wall (4 dB)
- Glass Wall with Metal Frame (6 dB)
- Heavy Door (15 dB)
- Light Door (4 dB)
- Metal Door (11 dB)
- Plasterboard Wall (3 dB)
- Thick Window (4 dB)
- Thin Window (2 dB)
- Window Office (3 dB)

Signal strength testing
Be aware of nearby interference that can impact your wireless access point installation

Physical network infrastructure
Topology
wireless network infrastructure, nodes are computers, smartphones, games consoles, printers and of course the router itself. Routers, hubs or switches are considered data communication equipment whereas data terminal equipment includes the devices that we typically interact with.

Understanding 'Airtime'
airtime describes how much time is available on a channel for data transmissions. Any time a device communicates with another device or an access point (AP) it uses airtime. Airtime Utilization is a per-channel statistic that defines what percentage of the channel is currently being used, and what percentage is therefore free. Airtime usage can come from:

Data traffic to and from client devices

802.11a/b/g/n/ac wireless protocol
802.11a: 54 Mbps standard, 5 GHz signaling (ratified 1999)
802.11b: 11 Mbps standard, 2.4 GHz signaling (1999)
802.11c: Operation of bridge connections (moved to 802.1D)
802.11d: Worldwide compliance with regulations for use of wireless signal spectrum (2001)
802.11e: Quality of Service support (2005) to improve the delivery of delay-sensitive applications, such as Voice Wireless LAN and streaming multimedia
802.11F: Inter-Access Point Protocol recommendation for communication between access points to support roaming clients (2003)
802.11g: 54 Mbps standard, 2.4 GHz signaling (2003)
802.11h: Enhanced version of 802.11a to support European regulatory requirements (2003)
802.11i: Security improvements for the 802.11 family (2004)
802.11j: Enhancements to 5 GHz signaling to support Japan regulatory requirements (2004)
802.11k: WLAN system management
802.11m: Maintenance of 802.11 family documentation
802.11n: 100+ Mbps standard improvements over 802.11g (2009)
802.11p: Wireless Access for the Vehicular Environment
802.11r: Fast roaming support using Basic Service Set transitions
802.11s: ESS mesh networking for access points
802.11T: Wireless Performance Prediction — recommendation for testing standards and metrics
802.11u: Internetworking with cellular and other forms of external networks
802.11v: Wireless network management and device configuration
802.11w: Protected Management Frames security enhancement

802.11y: Contention-Based Protocol for interference avoidance
802.11ac: 3.46Gbps standard, supports 2.4 and 5GHz frequencies through 802.11n

802.11 standards features overview
Bands, channels (frequencies) and channel widths

https://en.wikipedia.org/wiki/IEEE_802.11

scan-list
by default value is all channels from selected band that are supported by card and allowed by the country and frequency-mode settings (this list can be seen in info). For default scan list in 5ghz band channels are taken with 20MHz step, in 5ghz-turbo band - with 40MHz step, for all other bands - with 5MHz step. If scan-list is specified manually, then all matching channels are taken. (Example: scan-list=default,5200-5245,2412-2427 - This will use the default value of scan list for current band, and add to it supported frequencies from 5200-5245 or 2412-2427 range.)
Since RouterOS v6.0 with Winbox or Webfig, for inputting of multiple frequencies, add each frequency or range of frequencies into separate multiple scan-lists. Using a comma to separate frequencies is no longer supported in Winbox/Webfig since v6.0.

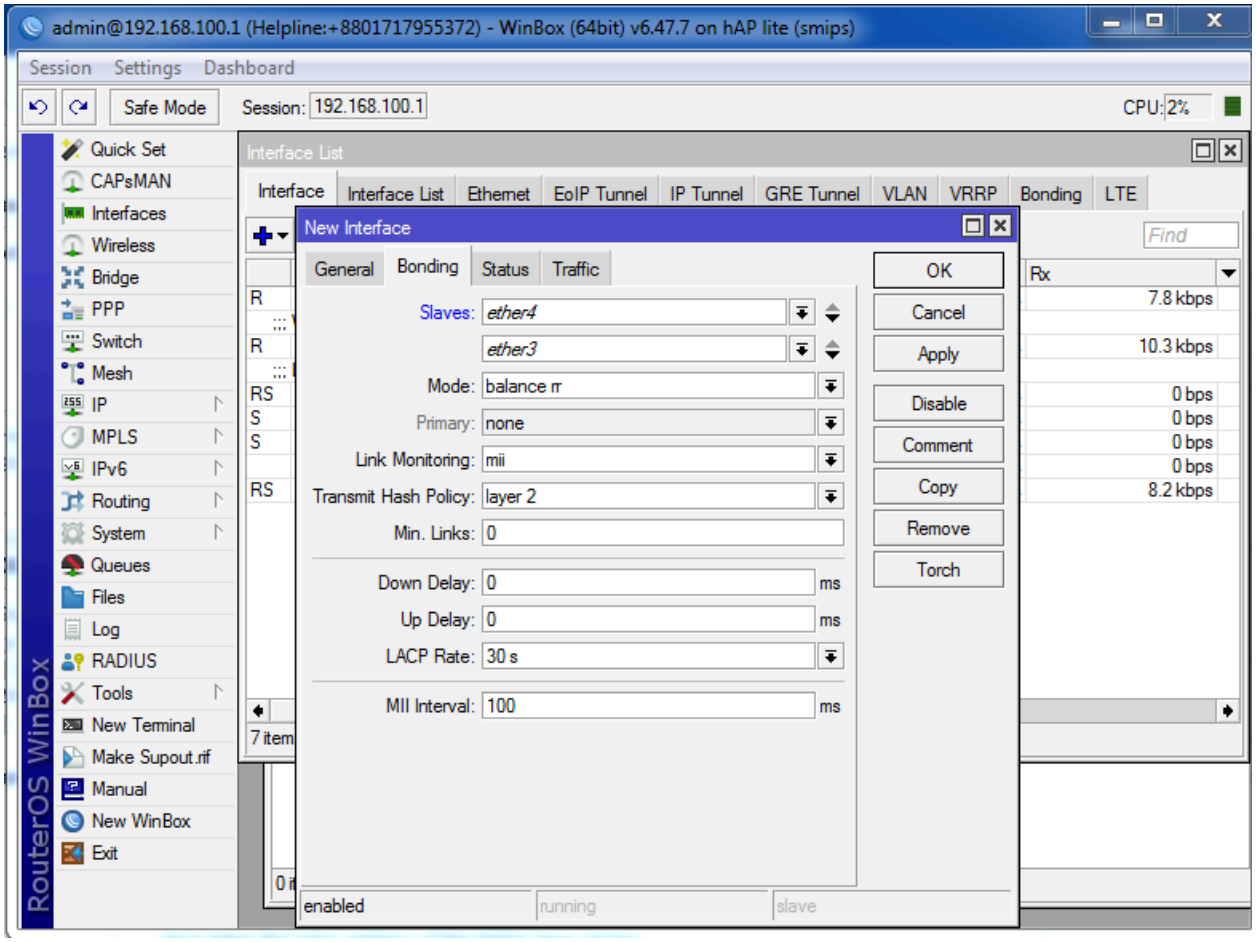
Since RouterOS v6.35 (wireless-rep) scan-list support step feature where it is possible to manually specify the scan step. Example: scan-list=5500-5600:20 will generate such scan-list values 5500,5520,5540,5560,5580,5600

Modulation schemes and MCS data rates
Modulation and Coding Scheme (MCS) Index Values can be used to determine the likely data rate of your WiFi connection. The MCS value essentially summarises the number of spatial streams, the modulation type and the coding rate that is possible when connecting your wireless access point.

MCS index	Spatial streams	Modulation type	Coding rate	Data rate (Mbit/s)			
				20 MHz channel		40 MHz channel	
				800 ns GI	400 ns GI	800 ns GI	400 ns GI
0	1	BPSK	1/2	6.50	7.20	13.50	15.00
1	1	QPSK	1/2	13.00	14.40	27.00	30.00
2	1	QPSK	3/4	19.50	21.70	40.50	45.00
3	1	16-QAM	1/2	26.00	28.90	54.00	60.00
4	1	16-QAM	3/4	39.00	43.30	81.00	90.00
5	1	64-QAM	2/3	52.00	57.80	108.00	120.00
6	1	64-QAM	3/4	58.50	65.00	121.50	135.00
7	1	64-QAM	5/6	65.00	72.20	135.00	150.00
8	2	BPSK	1/2	13.00	14.40	27.00	30.00
9	2	QPSK	1/2	26.00	28.90	54.00	60.00
10	2	QPSK	3/4	39.00	43.30	81.00	90.00
11	2	16-QAM	1/2	52.00	57.80	108.00	120.00
12	2	16-QAM	3/4	78.00	86.70	162.00	180.00
13	2	64-QAM	2/3	104.00	115.60	216.00	240.00
14	2	64-QAM	3/4	117.00	130.00	243.00	270.00
15	2	64-QAM	5/6	130.00	144.40	270.00	300.00
16	3	BPSK	1/2	19.50	21.70	40.50	45.00
17	3	QPSK	1/2	39.00	43.30	81.00	90.00
18	3	QPSK	3/4	58.50	65.00	121.50	135.00
19	3	16-QAM	1/2	78.00	86.70	162.00	180.00
20	3	16-QAM	3/4	117.00	130.00	243.00	270.00
21	3	64-QAM	2/3	156.00	173.30	324.00	360.00
22	3	64-QAM	3/4	175.50	195.00	364.50	405.00
23	3	64-QAM	5/6	195.00	216.70	405.00	450.00
24	4	BPSK	1/2	26.00	28.80	54.00	60.00
25	4	QPSK	1/2	52.00	57.60	108.00	120.00
26	4	QPSK	3/4	78.00	86.80	162.00	180.00
27	4	16-QAM	1/2	104.00	115.60	216.00	240.00
28	4	16-QAM	3/4	156.00	173.20	324.00	360.00
29	4	64-QAM	2/3	208.00	231.20	432.00	480.00
30	4	64-QAM	3/4	234.00	260.00	486.00	540.00
31	4	64-QAM	5/6	260.00	288.80	540.00	600.00

Channel bonding

Bonding is a technology that allows aggregation of multiple ethernet-like interfaces into a single virtual link, thus getting higher data rates and providing failover.



Chains (SISO, MIMO and MU-MIMO)

Therefore the different forms of single / multiple antenna links are defined as below:

SISO - Single Input Single Output

SIMO - Single Input Multiple output

MISO - Multiple Input Single Output

MIMO - Multiple Input multiple Output

MU-MIMO-Multi-user MIMO

CSMA/CA overview

Carrier-sense multiple access with collision avoidance (CSMA/CA) in computer networking, is a network multiple access method in which carrier sensing is used, but nodes attempt to avoid collisions by beginning transmission only after the channel is sensed to be "idle".[1][2] When they do transmit, nodes transmit their packet data in its entirety.

It is particularly important for wireless networks, where the collision detection of the alternative CSMA/CD is not possible due to wireless transmitters desensing their receivers during packet transmission.

HW protection (RTS/CTS)
Frame protection support (RTS/CTS)
802.11 standard provides means to protect transmission against other device transmission by using RTS/CTS protocol. Frame protection helps to fight the "hidden node" problem. There are several types of protection:

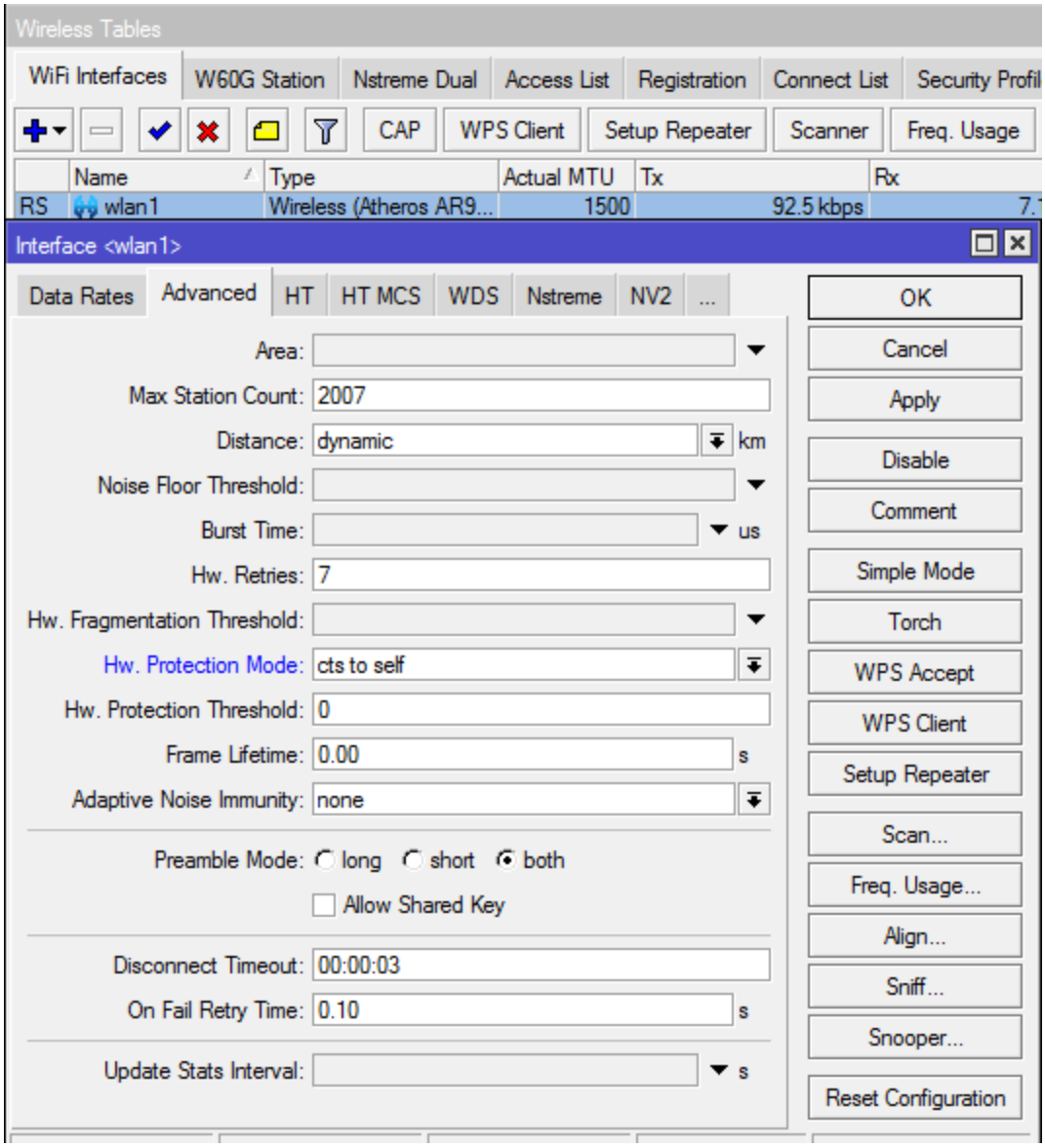
RTS/CTS based protection - device willing to send frame at first sends RequestToSend frame and waits for ClearToSend frame from intended destination. By "seeing" RTS or CTS frame 802.11 compliant devices know that somebody is about to transmit and therefore do not initiate transmission themselves
"CTS to self" based protection - device willing to send frame sends CTS frame "to itself". As in RTS/CTS protocol every 802.11 compliant device receiving this frame know not to transmit.
"CTS to self" based protection has less overhead, but it must be taken into account that this only protects against devices receiving CTS frame (e.g. if there are 2 "hidden" stations, there is no use for them to use "CTS to self" protection, because they will not be able to receive CTS sent by other station - in this case stations must use RTS/CTS so that other station knows not to transmit by seeing CTS transmitted by AP).
Protection mode is controlled by hw-protection-mode setting of wireless interface. Possible values: none - for no protection (default), rts-cts for RTS/CTS based protection or cts-to-self for "CTS to self" based protection.

Frame size threshold at which protection should be used is controlled by hw-protection-threshold setting of wireless interface.

For example, to enable "CTS-to-self" based frame protection on AP for all frames, not depending on size, use command:

[admin@MikroTik] /interface wireless> set 0 hw-protection-mode=cts-to-self hw-protection-threshold=0
To enable RTS/CTS based protection on client use command:

[admin@MikroTik] /interface wireless> set 0 hw-protection-mode=rts-cts hw-protection-threshold=0



mikrotik QoS priorities / WMM

Wi-Fi Multimedia

Wi-Fi Multimedia (WMM), previously known as Wireless Multimedia Extensions (WME), is a subset of the 802.11e wireless LAN (WLAN) specification that enhances quality of service (QoS) on a network by

prioritizing data packets according to four categories.

– Voice

– Video

– Best effort

– Background

How to set priority

Priority of packets can be set using "set priority" action of ip firewall mangle rules and/or bridge firewall filter rules. Priority can be set to specific value or to "ingress priority". Ingress priority is priority value that was detected on incoming packet, if available. Currently there are 2 sources of ingress priority - priority in VLAN header and priority from WMM packets received over wireless interface. For all other packets ingress priority is 0.

https://mum.mikrotik.com/presentations/ID17/presentation_4922_1510103908.pdf

Future standards (802.11ax)
The next-generation 802.11ax Wi-Fi standard, also known as Wi-Fi 6, is the latest step in a journey of nonstop innovation. The standard builds on the strengths of 802.11ac, adding efficiency, flexibility, and scalability that allows new and existing networks increased speed and capacity with next-generation applications.

The Institute of Electrical and Electronics Engineers (IEEE) proposed the 802.11ax standard so that it can couple the freedom and high speed of Gigabit Ethernet wireless with the reliability and predictability found in licensed radio.

802.11ax will let access points support more clients in dense environments and will provide a better experience for typical wireless LAN networks. It will also provide a more predictable performance for advanced applications such as 4K or 8K video, high-density high-definition collaboration apps, all-wireless offices, and the Internet of Things (IoT). 802.11ax will drive Wi-Fi toward the future as the growth of wireless continues.

Antenna gain and control of maximum EIRP
an antenna's power gain or simply gain is a key performance number which combines the antenna's directivity and electrical efficiency. In a transmitting antenna, the gain describes how well the antenna converts input power into radio waves headed in a specified direction. effective isotropic radiated power (EIRP). Effective isotropic radiated power is the hypothetical power that would have to be radiated by an isotropic antenna to give the same ("equivalent") signal strength as the actual source antenna in the direction of the antenna's strongest beam. The difference between EIRP and ERP is that ERP compares the actual antenna to a half-wave dipole antenna, while EIRP compares it to a theoretical isotropic antenna

Setting antenna gain on CAP
If you want to increase the Gain value, you change it locally to the wireless config of the Antenna and then you enable the CAP on the device...
Capsman will then get that Gain value

Selecting the country code and purpose of 'installation' setting
You must follow to regulatory domain requirements in your country. If you are allowed to use other frequencies, note that Antenna Gain and Transmit Power may decrease depending on

board and frequency. Devices are calibrated only for regulatory frequencies, use non standard frequencies at your own risk. The list only specifies frequencies accepted by the wireless chip, these frequencies might not always work due to antenna that is built into the product, device design, filters and other factors. USE STRICTLY AT YOUR OWN RISK

three frequency modes are available:
regulatory-domain - Limit available channels and maximum transmit power for each channel according to the value of country
manual-txpower - Same as above, but do not limit maximum transmit power.
superchannel - Conformance Testing Mode. Allow all channels supported by the card.

Dynamic frequency selection (DFS radar detect)
“Dynamic Frequency Selection” - not the infamous furniture company (although, it could be said this particular DFS “bargain sale” has finally ended with MikroTik!)

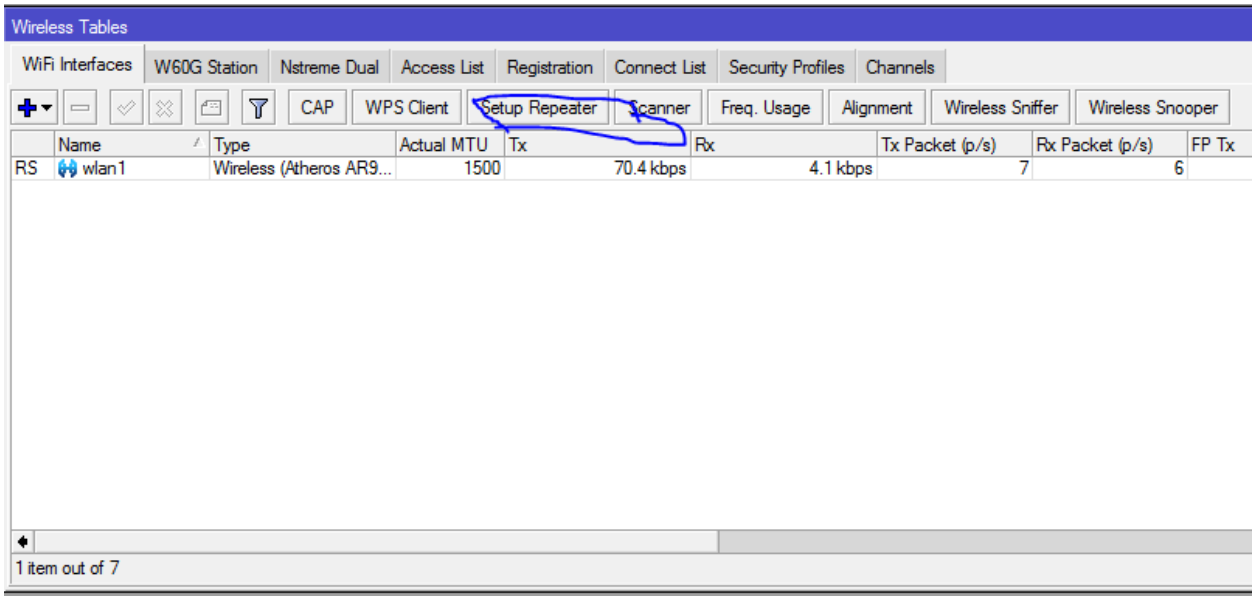
DFS - How does it work?
• Two main modes of detection:
– CAC - Channel Availability Check.
• A check done before operation on any channel
– ISM - In-Service Monitoring
• A check made constantly all the time while actually operating on any channel

CAC - Channel Availability Check.
• Before transmitting on any new channel, radio must monitor the channel for 60 seconds.
• On 5600-5650 this channel availability check is extended to 10 minutes
• If no Radar is detected, operation can begin and ISM mode is started.

How does MikroTik implement DFS from v6.37 onwards?
– On power up, initial CAC is performed on the frequency set into the frequency field and if that detects Radar, it then starts with the scan list

– DFS checks are only performed in ‘bridge’ & ‘ap-bridge’ modes not any ‘station’ modes

Extending coverage with repeaters and extenders



Bridging with MikroTik 60GHz Wireless Wire products

802.11ad implementation capable of providing Gigabit Ethernet speeds over wireless network.

Extend your Gigabit network over a transparent AES encrypted wireless 60GHz link without usual wired or wireless network problems.

/interface bridge
add name=bridge1
/interface bridge port
add bridge=bridge1 interface=ether1
add bridge=bridge1 interface=wlan60-1
/ip address
add address=192.168.88.1/24 interface=bridge1

mikrotik wireless Authentication (Open / Shared)
Without password

Encryption (WEP, WPA™ TKIP, WPA2™ AES)
Performance difference of TKIP vs. AES
TKIP and AES are two different types of encryption that can be used by a Wi-Fi network. TKIP is actually an older encryption protocol introduced with WPA to replace the very-insecure WEP encryption at the time. TKIP(Temporal Key Integrity Protocol) is actually quite similar to WEP encryption. TKIP is no longer considered secure, and is now deprecated. In other words, you shouldn't be using it.
AES(Advanced Encryption Standard) is a more secure encryption protocol introduced with WPA2.

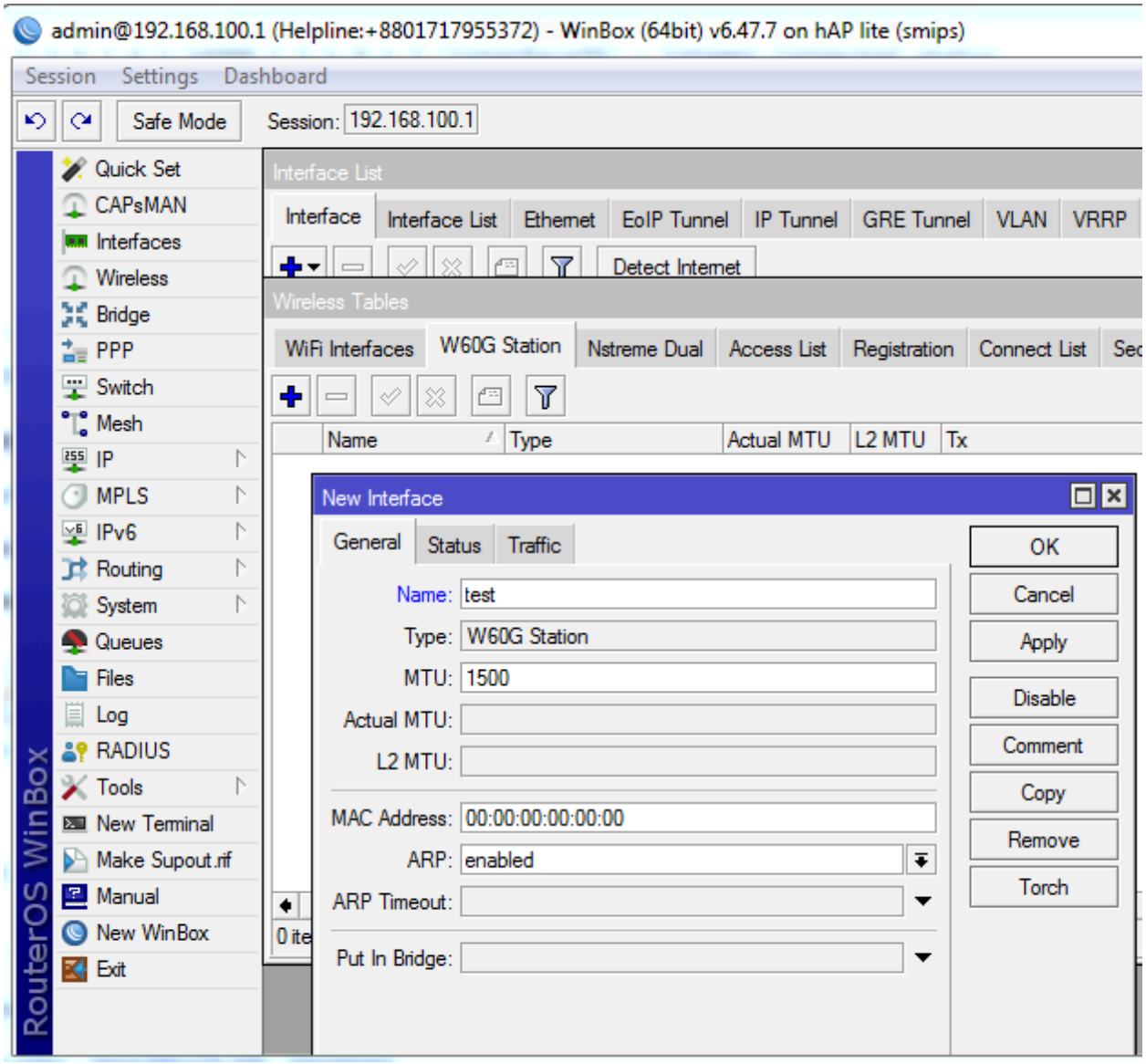
WPA
The Wi-Fi Alliance intended WPA as an intermediate measure to take the place of WEP pending the availability of the full IEEE 802.11i standard. WPA could be implemented through firmware upgrades on wireless network interface cards designed for WEP that began shipping as far back as 1999. However, since the changes required in the wireless access points (APs) were more extensive than those needed on the network cards, most pre-2003 APs could not be upgraded to support WPA.

The WPA protocol implements much of the IEEE 802.11i standard. Specifically, the Temporal Key Integrity Protocol (TKIP) was adopted for WPA. WEP used a 64-bit or 128-bit encryption key that must be manually entered on wireless access points and devices and does not change. TKIP employs a per-packet key, meaning that it dynamically generates a new 128-bit key for each packet and thus prevents the types of attacks that compromised WEP.

WPA2
Ratified in 2004, WPA2 replaced WPA. WPA2, which requires testing and certification by the Wi-Fi Alliance, implements the mandatory elements of IEEE 802.11i. In particular, it includes mandatory support for CCMP, an AES-based encryption mode.[7] [8] Certification began in September, 2004. From March 13, 2006 to June 30, 2020, WPA2 certification was mandatory for all new devices to bear the Wi-Fi trademark.[9][10]

WPA3
In January 2018, the Wi-Fi Alliance announced WPA3 as a replacement to WPA2.[11][12] Certification began in June, 2018.

The new standard uses an equivalent 192-bit cryptographic strength in WPA3-Enterprise mode[14] (AES-256 in GCM mode with SHA-384 as HMAC), and still mandates the use of CCMP-128 (AES-128 in CCM mode) as the minimum encryption algorithm in WPA3-Personal mode.



Overview of 802.11X (RADIUS and EAP)

802.1X is a port access protocol for protecting networks via authentication. As a result, this type of authentication method is extremely useful in the Wi-Fi environment due to the nature of the medium. If a Wi-Fi user is authenticated via 802.1X for network access, a virtual port is opened on the access point allowing for communication. If not successfully authorized, a virtual port isn't made available and communications are blocked.

There are three basic pieces to 802.1X authentication:

Supplicant A software client running on the Wi-Fi workstation.
Authenticator The Wi-Fi access point.

Authentication Server An authentication database, usually a radius server such as Cisco ACS*, Funk Steel-Belted RADIUS*, or Microsoft IAS*.

Basic access list (ACL) management

Access-List is a white list that indicates which MAC or IP can negotiate with specific port for connecting
if you need to deny specific MAC to connecting to your device bridge firewall is the option. just drop packets which are coming from specific MAC
if you need ACL in wireless devices , there is a built-in ACL in mikrotik wireless , but you should uncheck default authenticate feature from your WLAN interface.

Mitigating against most common known vulnerabilities of 802.11
Common Wireless Network Threats
1. Configuration Problems: Misconfigurations, incomplete configurations.

2. Denial of Service: Sending large amounts of traffic (or viruses) over the network with the intent of hijacking resources or introducing backdoors.

3. Passive Capturing: Eavesdropping within range of an access point to capture sensitive information.

4. Rogue (or Unauthorized/Ad-Hoc) Access Points: Fool devices into connecting with a false access point.

5. Evil Twin Attacks: Impersonating legit access points with a stronger signal to entice authorized users to sign on.

6. Hacking of Lost or Stolen Wireless Devices: Bypassing the password to gain access.

7. Freeloading: Piggybacking on a connection or intercepting file sharing.

General Tips for Minimizing Your Wireless Network's Security Threats
1. Implore your employees to use strong passwords (try a password generator).

2. Restrict access to only authorized users. For times when a guest requires access, set up a guest account with specific guest privileges on a separate wireless channel (and with a unique password) to maintain privacy of primary credentials.

3. Protect your SSID by renaming it, changing its default values, and disabling its broadcast to other users.

4. Encrypt your data with WPA3, which is currently the strongest encryption.

5. Install a host-based firewall.

6. Caution against improper file sharing:

Never share files over public networks.

Create a dedicated and restricted sharing directory.

Don’t open an entire hard drive for sharing.

Disable file sharing when it’s not needed.

7. Periodically update and install patches for all wireless access points.

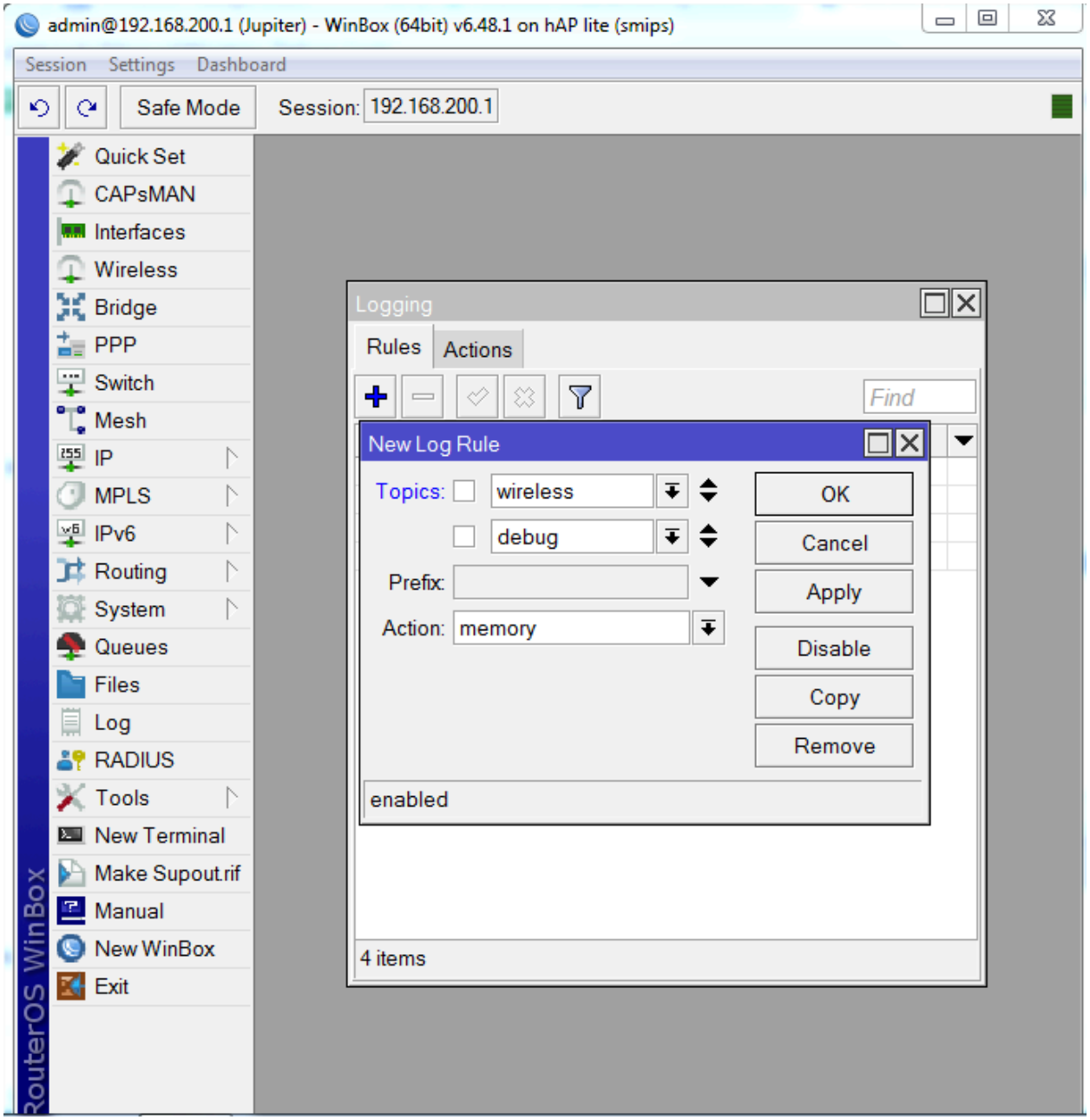
8. Use a VPN (Virtual Private Network) to allow out-of-office network connectivity among employees.

Troubleshooting wireless clients
22:33:20 wireless,debug wlan1: 00:80:48:41:AF:2A attempts to connect
22:33:20 wireless,debug wlan1: 00:80:48:41:AF:2A not in local ACL, by default accept
22:33:20 wireless,info 00:80:48:41:AF:2A@wlan1: connected

Debug Logs will give you more specific informantion on each step of the Client wireless connection and disconnection. The first line shows that the wireless client tried to connect to the AP. On the second line the AP checked to see if that client is allowed to connect to the AP and the resulting action. And only on the third line do you see that the client is connected. This is merely one example of the debug log messages. The description of all debug entries is written below.

To enable the wireless debug logs you should execute such commands:

[admin@MikroTik] > /system logging
[admin@MikroTik] system logging> add topics=wireless,debug action=memory



Registration table analysis
/interface wireless registration-table

In the registration table, you can see various information about currently connected clients. It is used only for Access Points.
All properties are read-only

TX/RX signal strength

Rx is received from the view of that interface.
Tx is transmitted from the view of that interface.
20db Best you can get
30 db VERY VERY GOOD signal
50 db very good signal
60 db good signal
70 db good signal but getting bad
80 db Not super but better than nothing
90 db Can you spell headache?
100 db R*I*P

Signal to noise ratio

Signal-to-noise ratio (SNR or S/N) is a measure used in science and engineering that compares the level of a desired signal to the level of background noise. SNR is defined as the ratio of signal power to the noise power, often expressed in decibels.
SNR calculations can be either simple or complex, and it depends on the devices in question and your available data. So, if your SNR measurements are already in decibel form, then you can subtract the noise quantity from the desired signal: SNR = S - N. This is because when you subtract logarithms, it is the equivalent of dividing normal numbers. Also, the difference in the numbers equals the SNR. For example, you measure a radio signal with a strength of -10 dB and a noise signal of -50 dB. -10 - (-50) = 40 dB.

5 dB to 10 dB: is below the minimum level to establish a connection, due to the noise level being nearly indistinguishable from the desired signal (useful information).

10 dB to 15 dB: is the accepted minimum to establish an unreliable connection.
15 dB to 25 dB: is typically considered the minimally acceptable level to establish poor connectivity.
25 dB to 40 dB: is deemed to be good.
41 dB or higher: is considered to be excellent.

CCQ, frames and HW frames, hardware retries
Client Connection Quality (CCQ) is a value in percent that shows how effective the bandwidth is used regarding the theoretically maximum available bandwidth.
The frames counts only those which contain actual data. In the case of Nstreme, only the ACK can be transmitted in a single frame, if there is no other data to send. These ACK frames will not be added to the frames count, but they will appear at hw-frames.

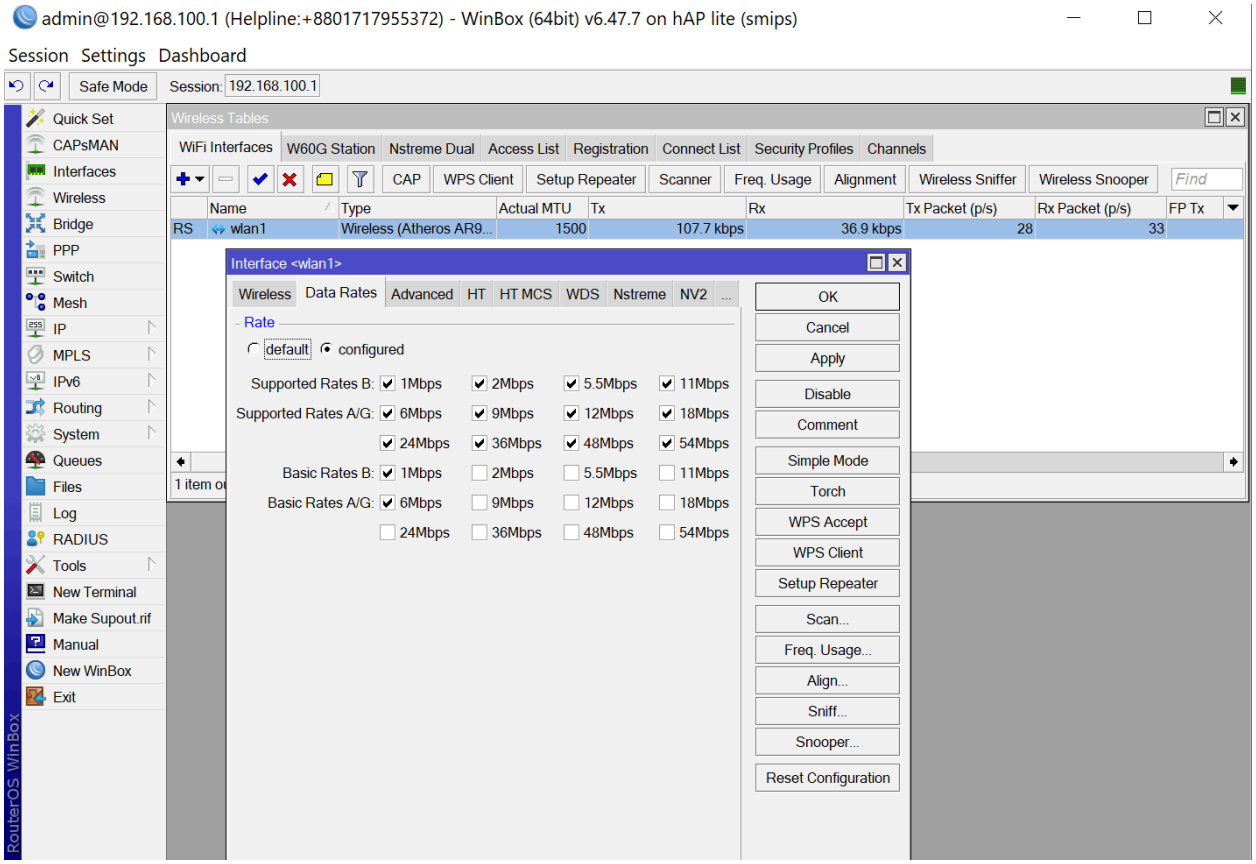
Number of times sending frame is retried without considering it a transmission failure. The data rate is decreased upon failure and frame is sent again. Three sequential failures on lowest

supported rate suspend transmission to this destination for the duration of on-fail-retry-time. After that, the frame is sent again. The frame is being retransmitted until transmission success, or until the client is disconnected after disconnect-timeout. The frame can be discarded during this time if frame-lifetime is exceeded. In case of Nstreme "on-fail-retry-time", "disconnect-timeout" and "frame-lifetime" settings are not used. So after three sequential failures on the lowest supported rate, the wireless link is disconnected with "extensive data loss" message.

Data rates
data rates or signalling rates are the theoretical speeds that a wireless access point can provide under perfect lab conditions, using all available spatial streams at the highest possible modulation rate.

IEEE 802.11 Wi-Fi protocol summary

Protocol	Frequency	Channel Width	MIMO	Maximum data rate (theoretical)
802.11ax	2.4 or 5GHz	20, 40, 80, 160MHz	Multi User (MU-MIMO)	2.4 Gbps[1]
802.11ac wave2	5 GHz	20, 40, 80, 160MHz	Multi User (MU-MIMO)	1.73 Gbps[2]
802.11ac wave1	5 GHz	20, 40, 80MHz	Single User (SU-MIMO)	866.7 Mbps[2]
802.11n	2.4 or 5 GHz	20, 40MHz	Single User (SU-MIMO)	450 Mbps[3]
802.11g	2.4 GHz	20 MHz	N/A	54 Mbps
802.11a	5 GHz	20 MHz	N/A	54 Mbps
802.11b	2.4 GHz	20 MHz	N/A	11 Mbps
Legacy 802.11	2.4 GHz	20 MHz	N/A	2 Mbps



Analysing the System log for wireless problems mikrotik
y default RouterOS wireless log shows that client connects and disconnects as simple entries:

22:32:18 wireless,info 00:80:48:41:AF:2A@wlan1: connected

It is enough for regular users to know that the wireless client with MAC address "00:80:48:41:AF:2A" is connected to wireless interface "wlan1". But actually there are more log entries available than are shown in standard logging. They are called 'debug' logs which give more detailed information. In the following Debug Log example you will see the same client connecting to the AP in more detail than found in typical logging:

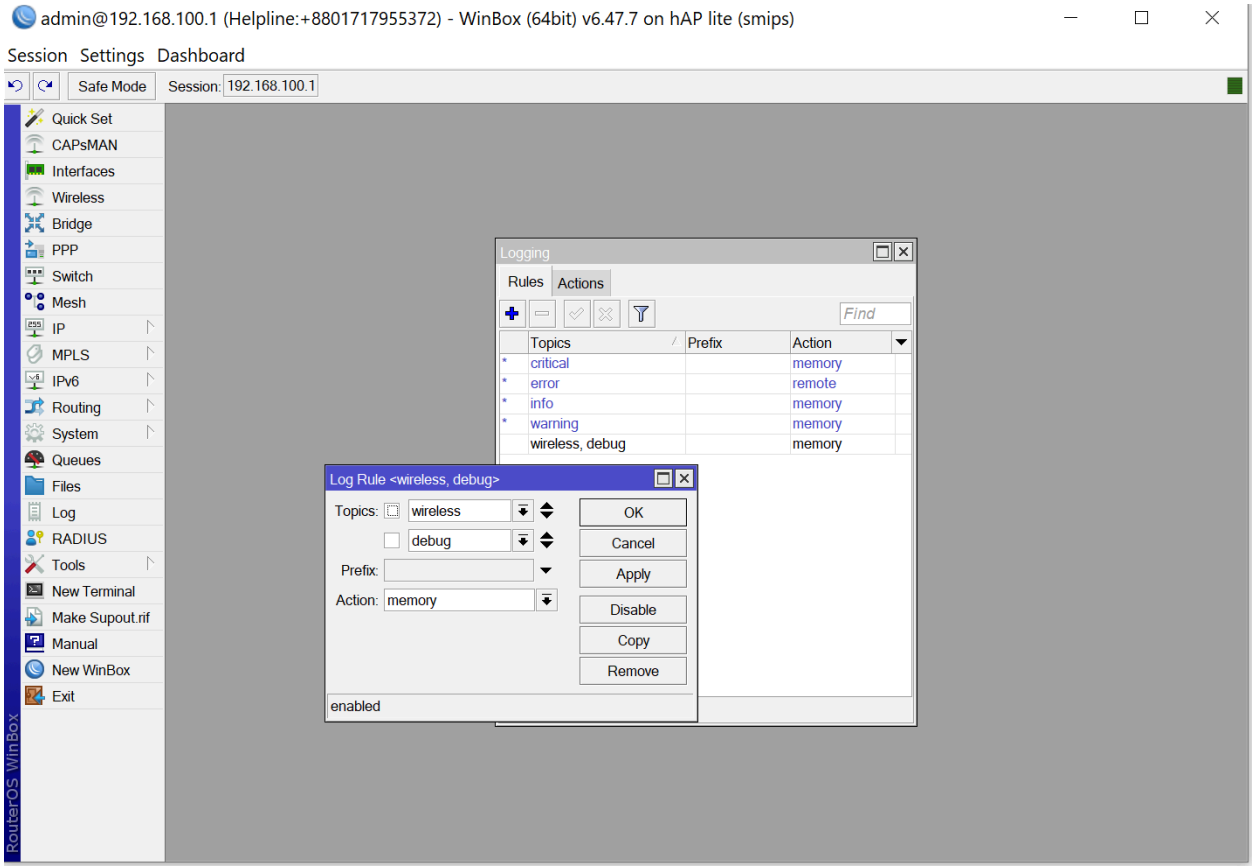
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22:33:20 wireless,debug wlan1: 00:80:48:41:AF:2A not in local ACL, by default accept
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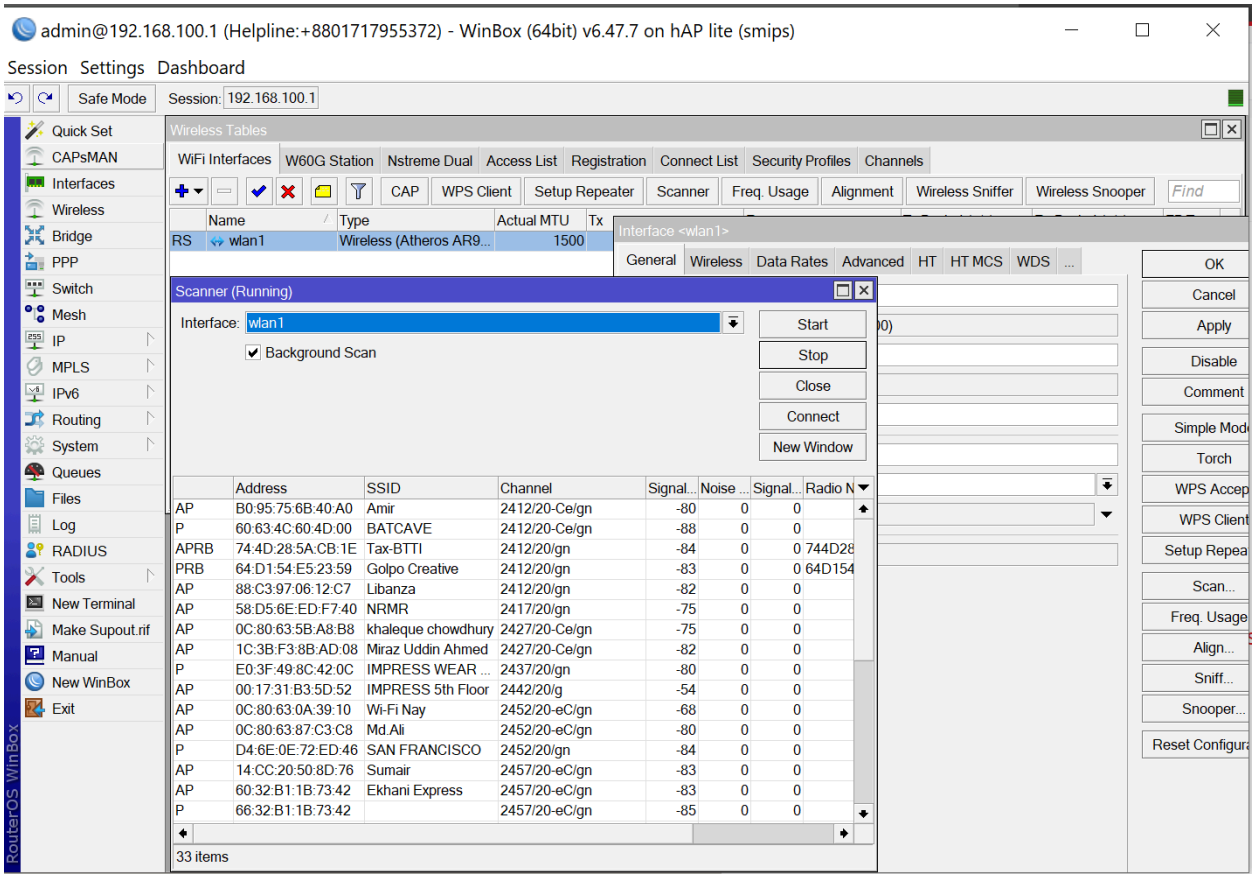
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[admin@MikroTik] > /system logging
[admin@MikroTik] system logging> add topics=wireless,debug action=memory



mikrotik wireless Scan, background scan

Beginning with Mikrotik version 6.35, the availability of Background Scan on the wireless interface. From the wiki: "The scan command allows you to see available AP's in the frequency range defined in the scan-list. Using the scan command the interface operation is disabled (wireless link is disconnected during the scan operation). Since RouterOS v6.35 (wireless-rep) background scan is supported which can be used during the wireless interface operation without disconnecting the wireless links. Background scan is supported only using the 802.11 wireless protocol." This is a great way to scan for other wireless networks without disrupting the registered clients. Notice in the screenshot, clients are connected but I am still scanning.

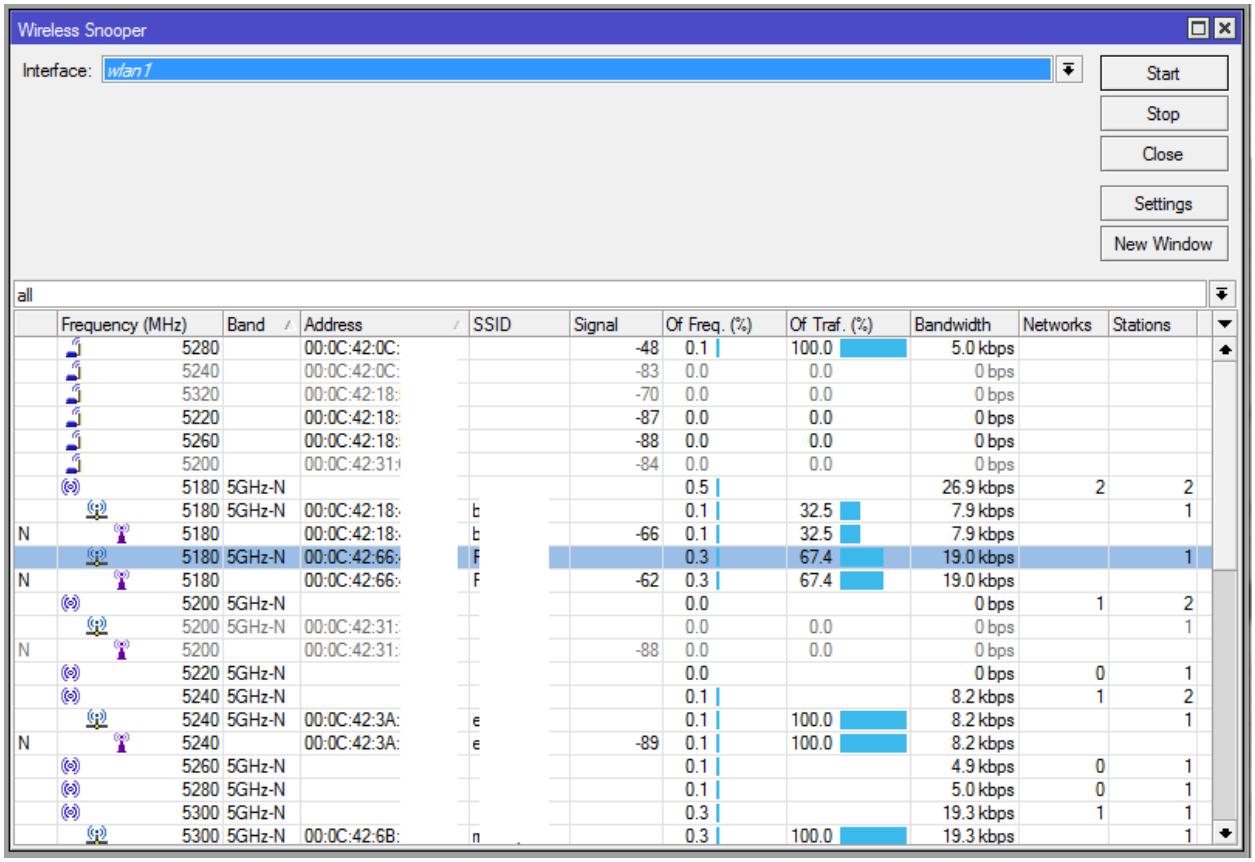


Frequency usage

If no country is set, these frequencies are used (FCC compliant set of channels)
2.4GHz mode: 2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462
2.4GHz-g-turbo mode: 2437
5GHz mode: 5180, 5200, 5220, 5240, 5260, 5280, 5300, 5320, 5745, 5765, 5785, 5805, 5825
5GHz-turbo mode: 5210, 5250, 5290, 5760, 5800

Wireless Snooper

This tool monitors surrounding frequency usage, and displays which devices occupy each frequency. It's available both in console, and also in Winbox.
/interface wireless snooper



Wireless sniffer

Wireless sniffer allows to capture frames including Radio header, 802.11 header and other wireless related information.
/interface wireless sniffer

Pre-installation wireless site survey
During pre-installation site surveys, 802 Networks will conduct on-site tests using the access point brand that your customer is planning to install. We use the standard brands Aerohive, Aruba, Cisco, Meraki and Commscope Ruckus for this purpose. Does your customer have something else in mind? We would be happy to take that up as well!

The procedure for pre-installation wireless surveys is as follows: we place the access point and measure the extent to which the coverage meets the required usage requirements. A detailed report of the site survey is then sent to you, containing the access point position, wireless coverage, RF interference and instructions for installing the access points.

Post-installation wireless site survey

Does your customer already have an installation but requires a few adjustments? A post-installation site survey measures the existing wireless network infrastructure and tests it against your customer's requirements.

A conclusive report will of course be provided to you containing the results, including the position of the access point, wireless coverage, RF interference and our recommendations.

Spectrum analysis overview
Why spectrum analysis?

Since RF signals are invisible to the naked eye, a spectrum analyzer is necessary to see into the wireless landscape to observe what is transmitting and where in the spectrum the "noise" is occurring. Sometimes the solution is to change the channel of the Wi-Fi network to avoid the other signals, and sometimes eliminating the offending wireless devices that "don't play well with others" is the answer. Occasionally, in situations where interference cannot be avoided or eliminated, the only solution is to switch Wi-Fi bands completely. Without spectrum analysis, implementing the proper solution is an expensive and time-consuming game of trial and error.

2.4 GHz Overview
The 2.4 GHz band contains eleven channels (and up to 13 or 14 in Europe and Japan, respectively), but only channels 1, 6, and 11 do not overlap. The 2.4 GHz band offers the widest compatibility with Wi-Fi devices through 802.11b/g/n, but with only three channels to chose from, the 2.4 GHz band is often very limited in the amount of traffic that it can sustain. While the 2.4 GHz band offers good range, it also suffers from non-Wi-Fi interference caused by electronic devices like cordless phones, security systems, microwave ovens, wireless audio-visual systems, Bluetooth devices, and more.

5 GHz Overview
The 5 GHz band contains 24 non-overlapping channels, which gives it a significant advantage over the 2.4 GHz band's three non-overlapping channels. With more channels to choose from, interference avoidance is as simple as choosing an unused channel. Additionally, fewer electronic devices use 5 GHz, meaning Wi-Fi access points operating in this frequency range have fewer RF signals to compete with. Although the 5 GHz band is relatively quiet and has many more available channels, it is widely believed to be only a matter of time before this band is as congested as the 2.4 GHz band.

Prediction software overview
TamoGraph is a powerful and user-friendly wireless site survey software tool for collecting, visualizing, and analyzing 802.11 a/b/g/n/ac/ax Wi-Fi data. Wireless network deployment and maintenance requires the use of a professional RF site survey tool that facilitates otherwise time-consuming and very complex tasks, such as ongoing analysis and reporting of signal strength, noise and interference, channel allocation, data rates, etc.

By using TamoGraph, businesses can dramatically reduce the time and costs that are involved in deploying and maintaining Wi-Fi networks and improve network performance and coverage in all kinds of environments: office buildings, airports, cafes, malls, or open space areas.

VisiWave Site Survey
Visualize Your Wireless Network™
Easily perform professional wireless site surveys. Analyze your WiFi network, create coverage heatmaps, see coverage voids, view channel usage/overlap, and more.

WiFi wireless networks are everywhere. They are a critical component of most business infrastructures. They need to be examined, explained, and held accountable. VisiWave Site Survey is the software tool that shows you what's going on inside your 802.11 network. VisiWave Site Survey provides advanced data collection and visualization capabilities that form a complete wireless LAN site survey solution that allows you to visualize the radio waves and demonstrate the effectiveness of your Wi-Fi coverage.

MikroTik CAPsMAN version 2 features
Controlled Access Point system Manager (CAPsMAN) allows centralization of wireless network management and if necessary, data processing. When using the CAPsMAN feature, the network will consist of a number of 'Controlled Access Points' (CAP) that provide wireless connectivity and a 'system Manager' (CAPsMAN) that manages the configuration of the APs, it also takes care of client authentication and optionally, data forwarding.

CAPsMAN v2 features:

CAPsMAN automatic upgrade of all CAP clients (configurable)
improved CAP<->CAPsMAN data connection protocol
added "Name Format" and "Name Prefix" setting for Provision rules
improved logging entries when client roams between the CAPs
added L2 Path MTU discovery

CAP hardware/software requirements
CAPsMAN works on any RouterOS device from v6.11, wireless interfaces are not required (since it manages the wireless interfaces of CAPs)
CAPsMAN v2 is working starting from RouterOS v6.22rc7.
CAPsMAN v1 is discontinued starting from 6.37.
CAP device should have at least Level4 RouterOS license

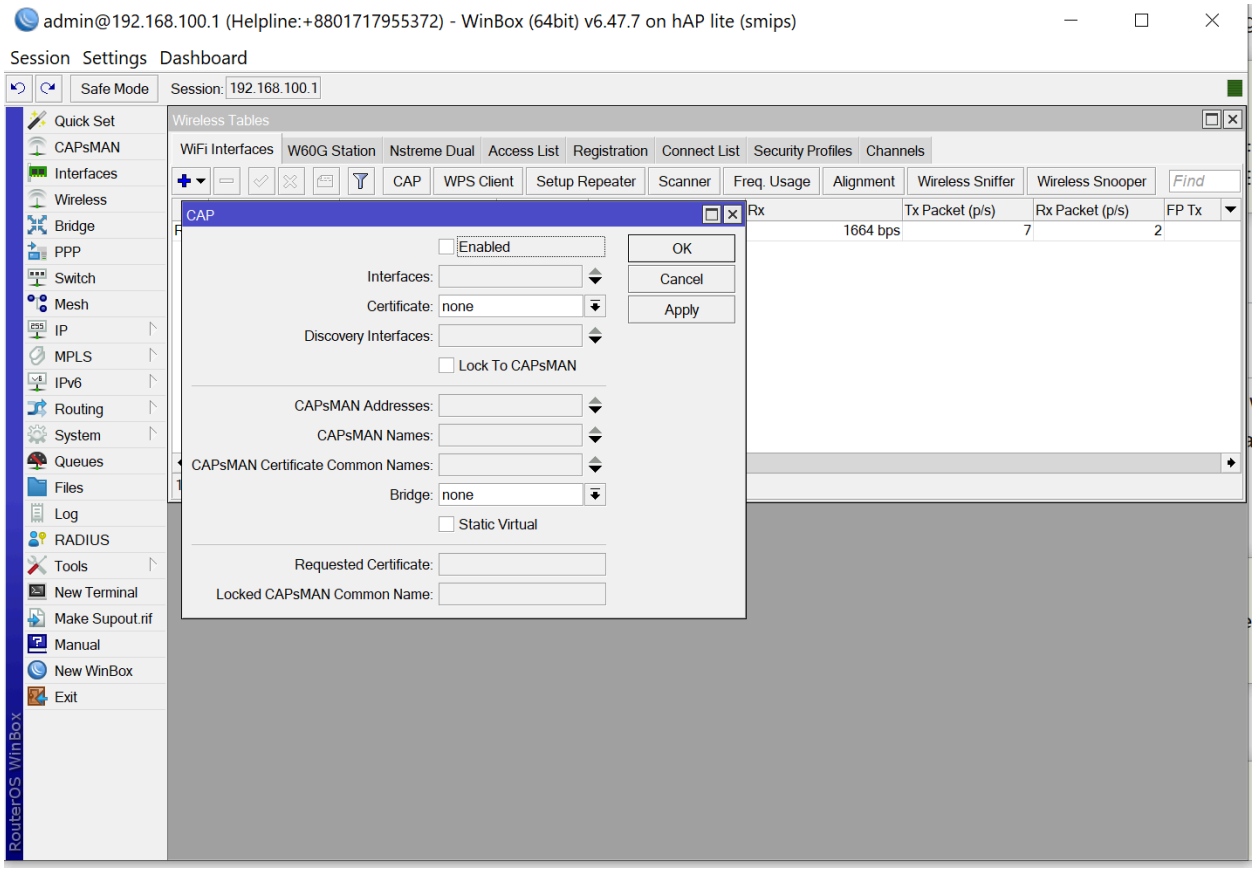
L2 (broadcast/multicast) vs L3 (via UDP) CAPs
CAP and CAPsMAN must be on the same Layer 2 segment - either physical or virtual (by means of L2 tunnels)

if the CAP is not on the same L2 segment as CAPsMAN, it must be provisioned with the CAPsMAN IP address, because IP multicast based discovery does not work over Layer3

IP layer (UDP) connection features:
can traverse NAT if necessary
CAP must be able to reach CAPsMAN using IP protocol
if the CAP is not on the same L2 segment as CAPsMAN, it must be provisioned with the CAPsMAN IP address, because IP multicast based discovery does not work over Layer3

Using DHCP option 138
Comma-separated list of IP addresses for one or more CAPsMAN system managers. DHCP Option 138 (capwap) will be used.

Configuration of a CAP



CAPsMAN discovery and selection by CAP
For the CAPsMAN system to function and provide wireless connectivity, a CAP must establish management connection with CAPsMAN. A management connection can be established using MAC or IP layer protocols and is secured using 'DTLS'.

A CAP can also pass the client data connection to the Manager, but the data connection is not secured. If this is deemed necessary, then other means of data security needs to be used, e.g. IPSec or encrypted tunnels.

CAP to CAPsMAN connection can be established using 2 transport protocols (via Layer 2 and Layer3).

MAC layer connection features:
no IP configuration necessary on CAP
CAP and CAPsMAN must be on the same Layer 2 segment - either physical or virtual (by means of L2 tunnels)
IP layer (UDP) connection features:
can traverse NAT if necessary
CAP must be able to reach CAPsMAN using IP protocol
if the CAP is not on the same L2 segment as CAPsMAN, it must be provisioned with the CAPsMAN IP address, because IP multicast based discovery does not work over Layer3
In order to establish connection with CAPsMAN, CAP executes a discovery process. During discovery, CAP attempts to contact CAPsMAN and builds an available CAPsMANs list. CAP attempts to contact to an available CAPsMAN using:

configured list of Manager IP addresses
list of CAPsMAN IP addresses obtained from DHCP server
broadcasting on configured interfaces using both - IP and MAC layer protocols.
When the list of available CAPsMANs is built, CAP selects a CAPsMAN based on the following rules:

if caps-man-names parameter specifies allowed manager names (/system identity of CAPsMAN), CAP will prefer the CAPsMAN that is earlier in the list, if list is empty it will connect to any available Manager
suitable Manager with MAC layer connectivity is preferred to Manager with IP connectivity
After Manager is selected, CAP attempts to establish DTLS connection. There are the following authentication modes possible:

no certificates on CAP and CAPsMAN - no authentication
only Manager is configured with certificate - CAP checks CAPsMAN certificate, but does not fail if it does not have appropriate trusted CA certificate, CAPsMAN must be configured with require-peer-certificate=no in order to establish connection with CAP that does not possess certificate
CAP and CAPsMAN are configured with certificates - mutual authentication
After DTLS connection is established, CAP can optionally check CommonName field of certificate provided by CAPsMAN. caps-man-certificate-common-names parameter contains list of allowed CommonName values. If this list is not empty, CAPsMAN must be configured with certificate. If this list is empty, CAP does not check CommonName field.

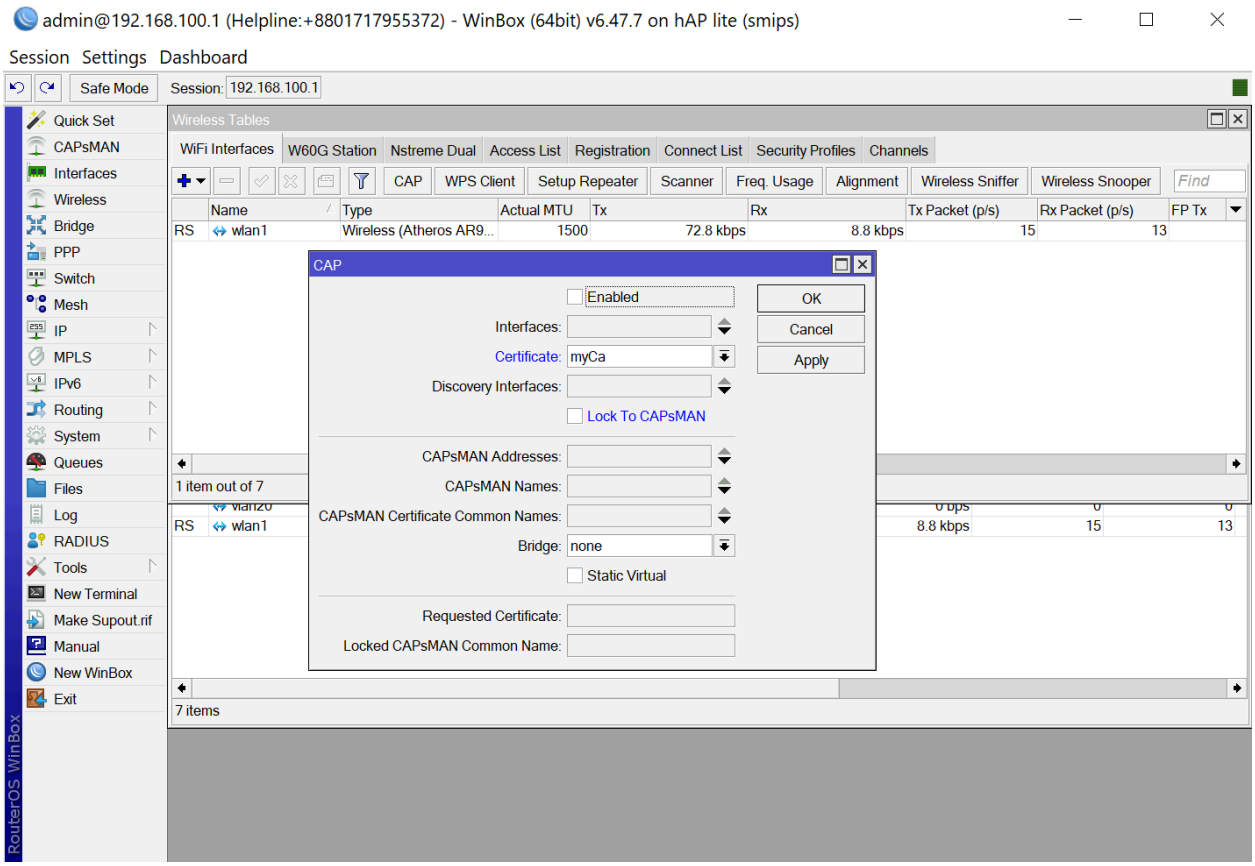
If the CAPsMAN or CAP gets disconnected from the network, the loss of connection between CAP and CAPsMAN will be detected in approximately 10-20 seconds.

Authentication and locking by SSL certificates
To simplify CAPsMAN and CAP configuration when certificates are required (e.g. for automatic locking feature), CAPsMAN can be configured to generate necessary certificates automatically and CAP can be configured to request certificate from CAPsMAN.

Automatic certificates do not provide full public key infrastructure and are provided for simple setups. If more complicated PKI is necessary - supporting proper certificate validity periods, multiple-level CA certificates, certificate renewal - other means must be used, such as manual certificate distribution or SCEP.

CAPsMAN has the following certificate settings:

certificate - this is CAPsMAN certificate, private key must be available for this certificate. If set to none, CAPsMAN will operate in no-certificate mode and none of certificate requiring features will work. If set to auto, CAPsMAN will attempt to issue certificate to itself using CA certificate (see ca-certificate description). Note that CommonName automatically issued certificate will be "CAPsMAN-<mac address>" and validity period for will be the same as for CA certificate.
ca-certificate - this is CA certificate that CAPsMAN will use when issuing certificate for itself if necessary (see certificate description) and when signing certificate requests from CAPs. If set to none, CAPsMAN will not be able to issue certificate to itself or sign certificate requests from CAPs. If set to auto, CAPsMAN will generate self-signed CA certificate to use as CA certificate. CommonName for this certificate will take form "CAPsMAN-CA-<mac address>" and validity period will be from jan/01/1970 until jan/18/2038.

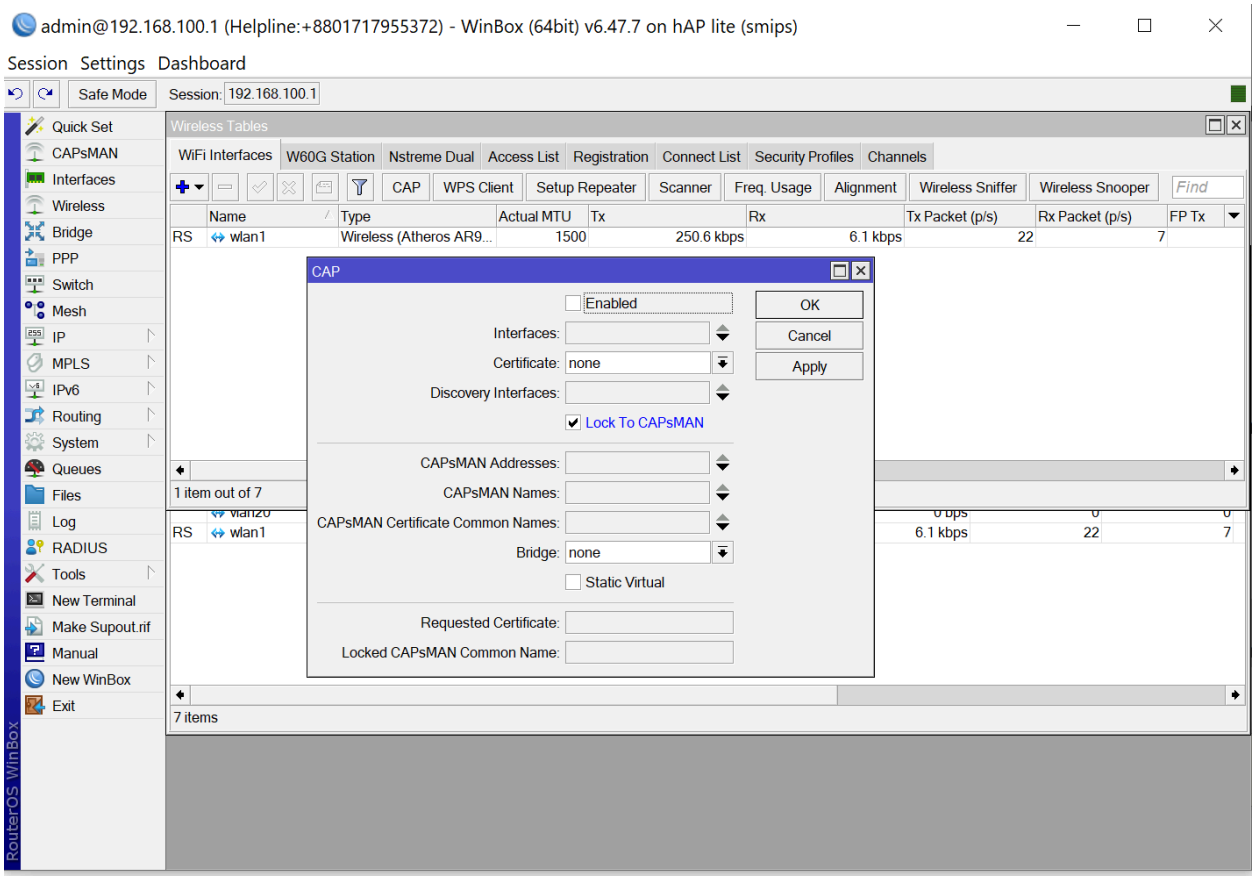


Auto certificate & locking

CAP can be configured to automatically lock to particular CAPsMAN. Locking is implemented by recording certificate CommonName of CAPsMAN that CAP is locked to and checking this CommonName for all subsequent connections. As this feature is implemented using certificate CommonName, use of certificates is mandatory for locking to work.

Locking is enabled by the following command:

[admin@CAP] > /interface wireless cap set lock-to-caps-man=yes



Auto upgrading feature

CAPsMAN v2 is NOT compatible with current CAPsMAN v1 (CAPsMAN v1 CAP devices will not be able to connect to CAPsMAN v2 and CAPsMAN v2 CAP devices will not be able to connect to CAPsMAN v1). It means that both CAPsMAN and CAP devices should have wireless-cm2 package enabled/installed in order to make CAPsMAN v2 system to work.

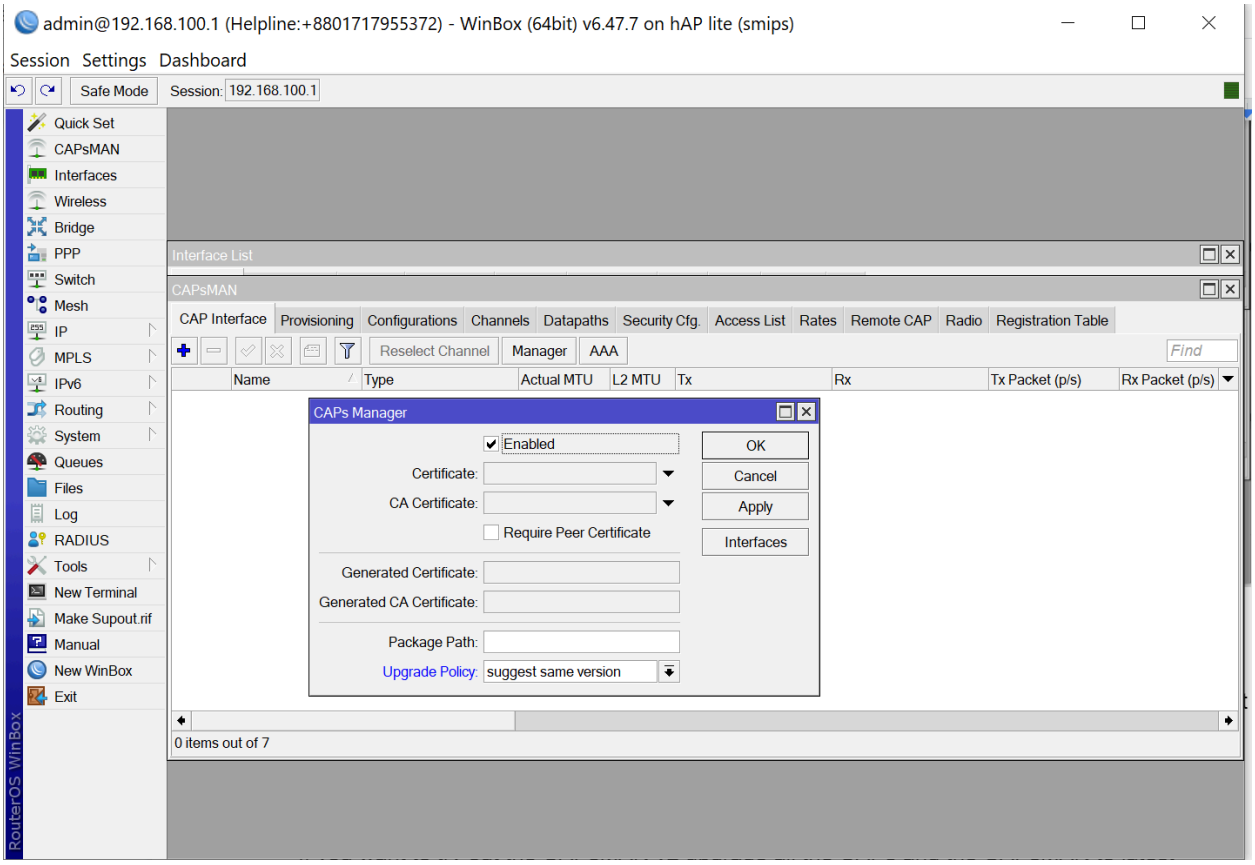
If you want to try out the CAPsMAN v2 upgrade all the CAPs and the CAPsMAN to latest RouterOS version and enable/install wireless-cm2 package.

CAPsMAN v2 features:

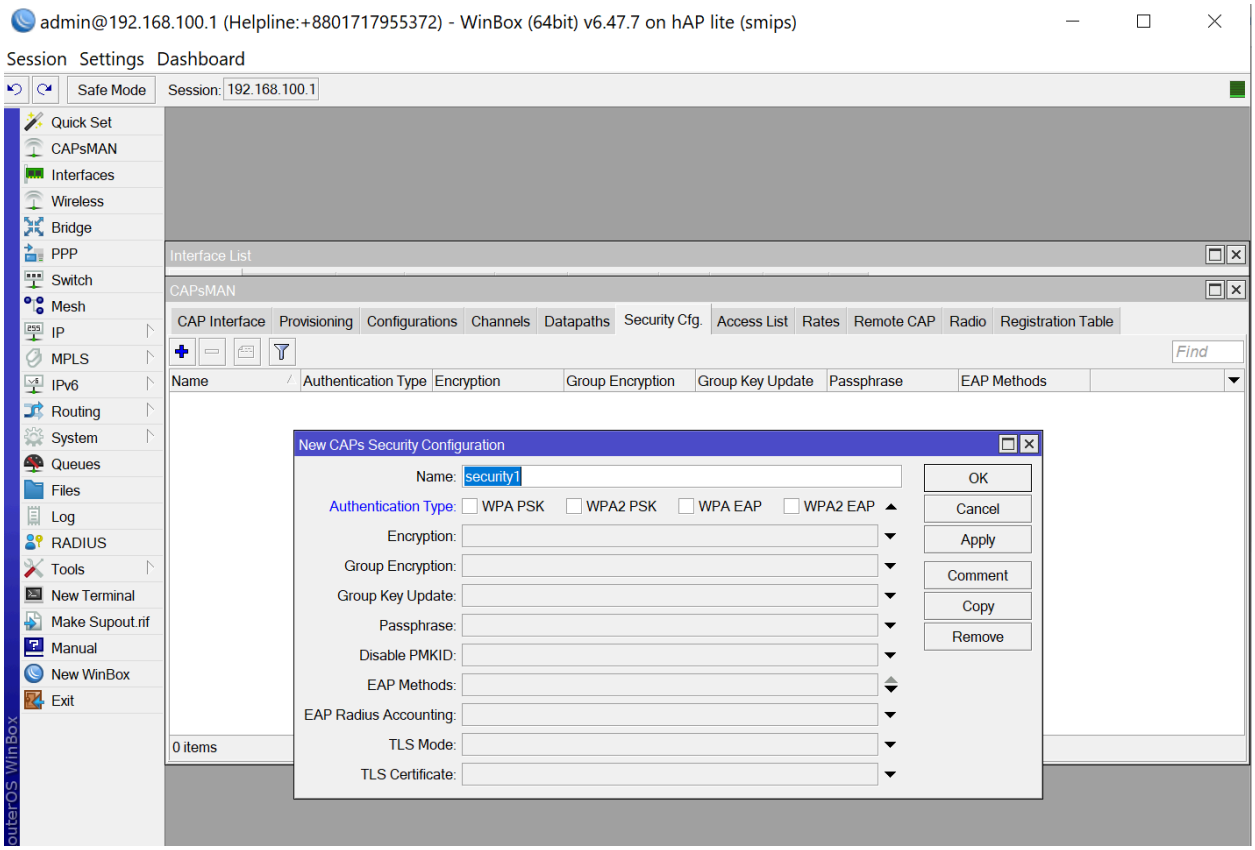
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improved CAP<->CAPsMAN data connection protocol
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added L2 Path MTU discovery
Upgrade options from v1 to v2:

Option1: Install a new temporary CAPsMAN v2 router in same network where the current CAPsMAN router is and start enabling/upgrading wireless-cm2 package on the CAPs. All CAPs with the v2 will connect to the new temporary CAPsMAN v2 router. After every CAP is upgraded to v2, upgrade your current CAPsMAN to v2 and then turn off the temporary CAPsMAN v2 router.

Option2: Upgrade your CAPs and then CAPsMAN to v2 at the same time. In this case you could have little more downtime unless you schedule all the CAPs to reboot/install at the same time.



Securing the CAP configuration



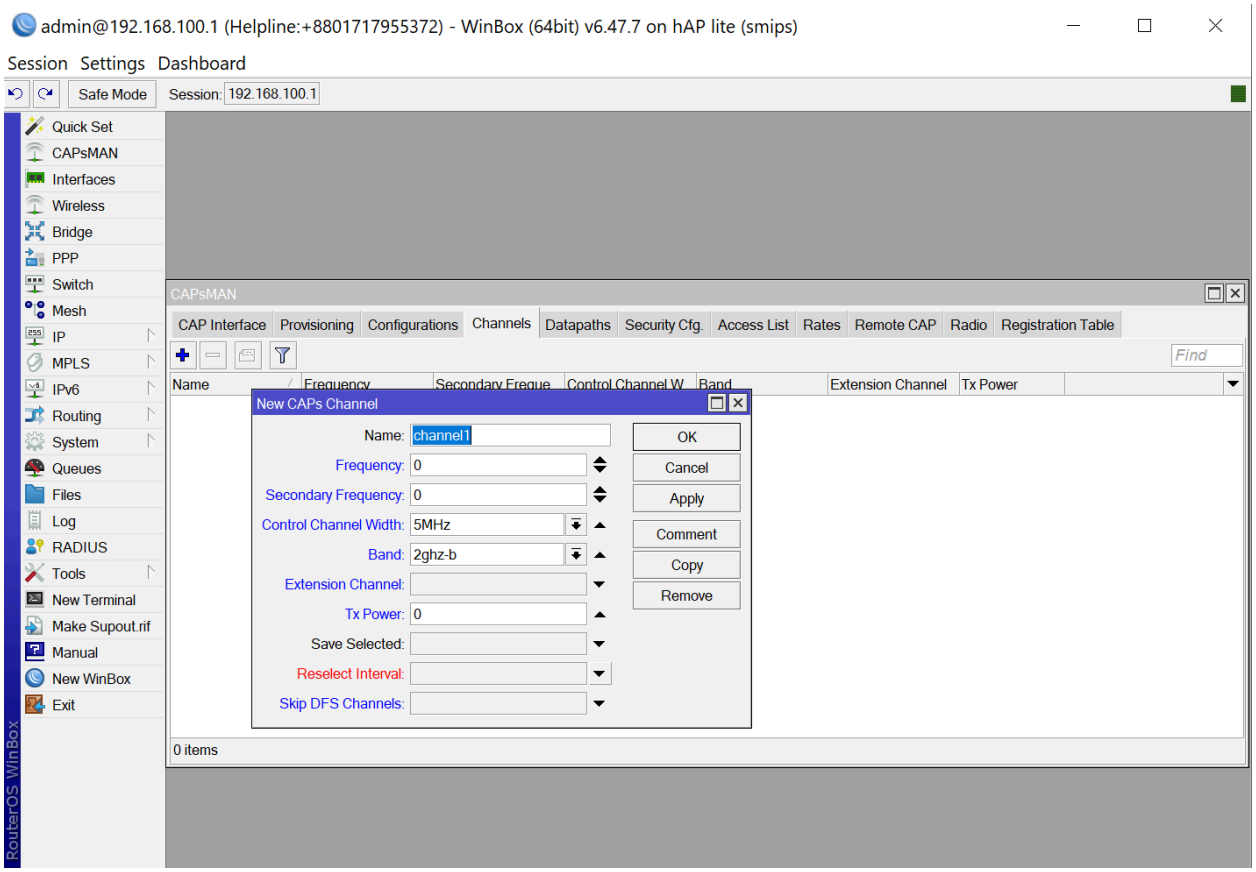
CAPsMAN configuration settings (channels, datapaths, security configurations, data rates)

Channel

Channel group settings allows for the configuration of lists of radio channel related settings, such as radio band, frequency, Tx Power extension channel and width.

Channel group settings are configured in the Channels profile menu /caps-man channels

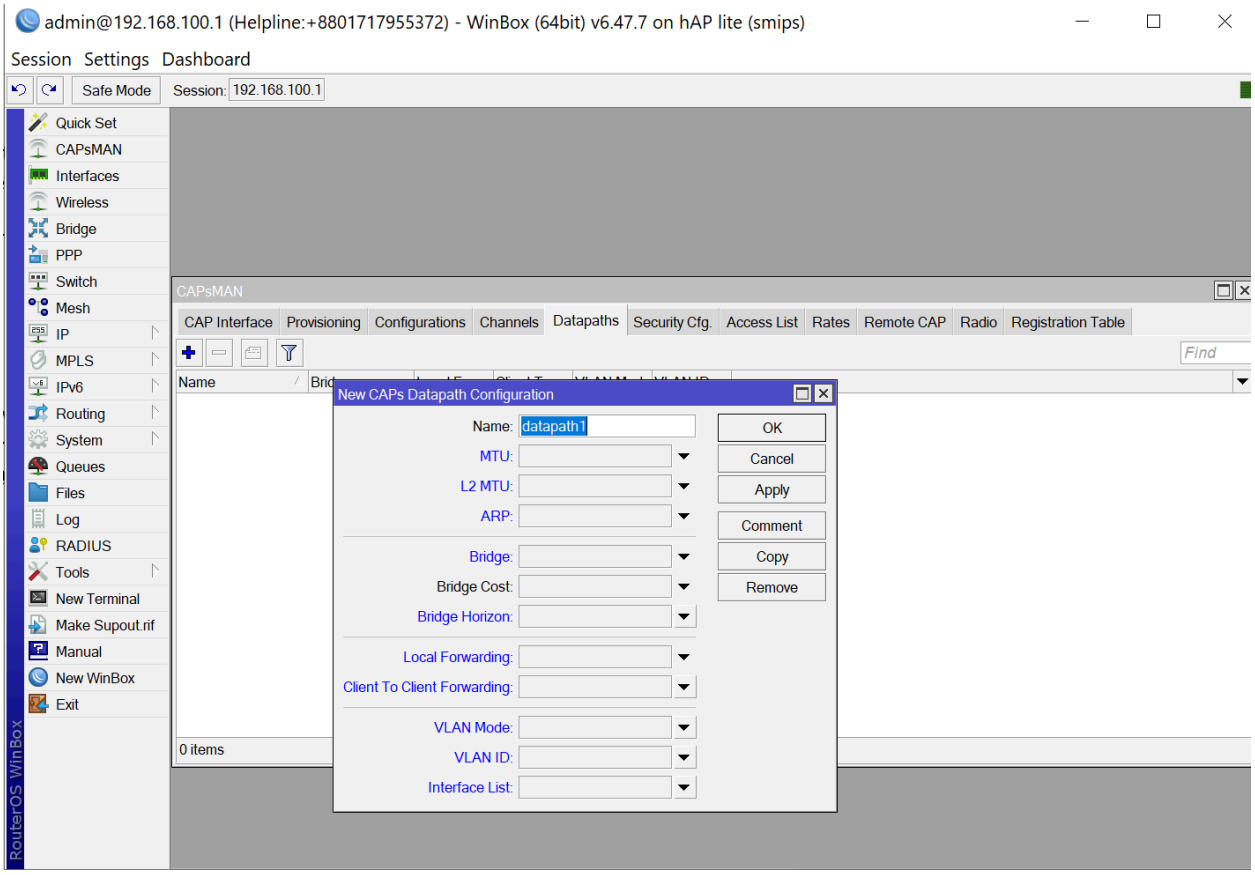
/caps-man channels



Datapath

Datapath settings control data forwarding related aspects. On CAPsMAN datapath settings are configured in datapath profile menu /caps-man datapath or directly in a configuration profile or interface menu as settings with datapath.

/caps-man datapath



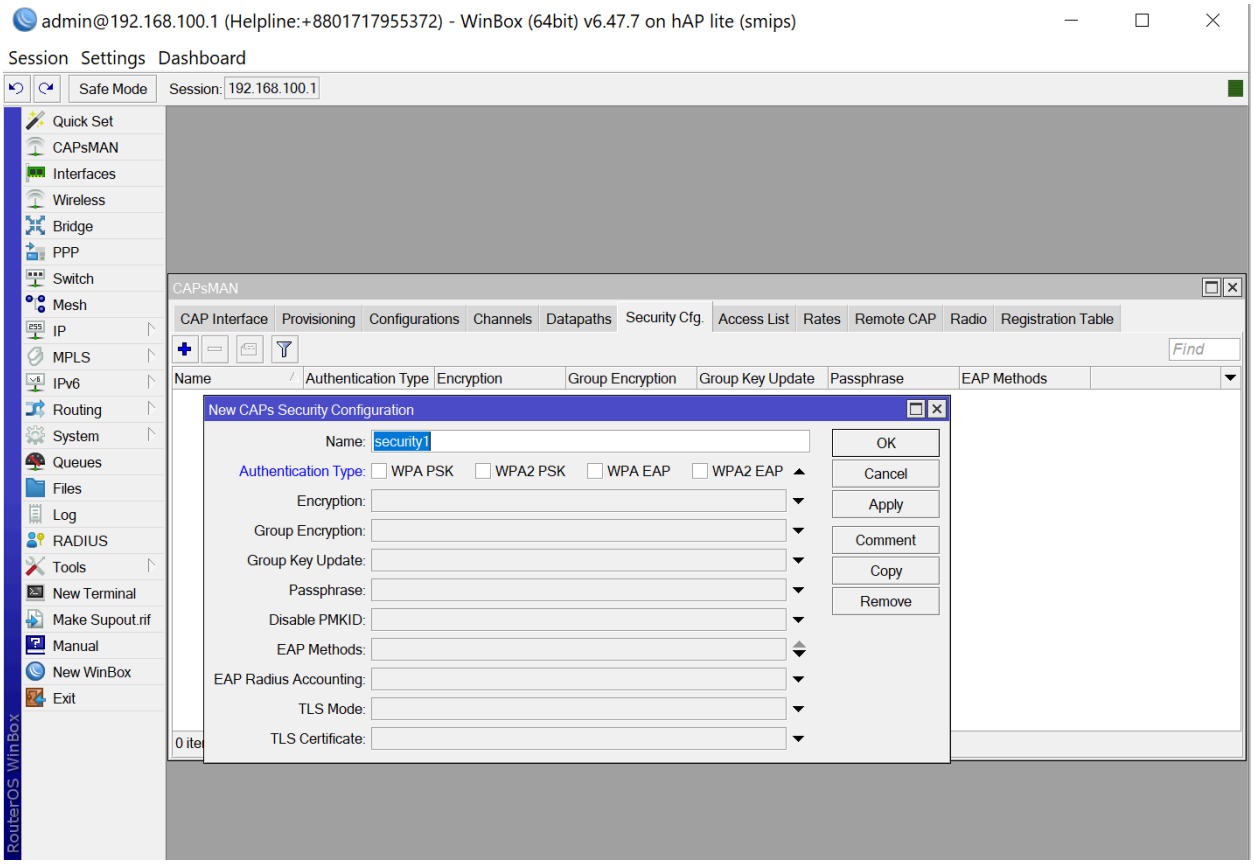
Security
Create security profile for WPA2 PSK, without specifying passphrase:

[admin@CM] /caps-man security>add name="wpa2psk" authentication-types=wpa2-psk encryption=aes-ccm
Create configuration profile to be used by master interface

specify WPA2 passphrase in configuration
specify channel settings in configuration:
[admin@CM] /caps-man configuration> add name=master-cfg ssid=master security=wpa2psk security.passphrase=12345678 channel.frequency=5180 channel.width=20 channel.band=5ghz-a
Create configuration profile to be used by virtual AP interface

specify different WPA2 passphrase in configuration:
[admin@CM] /caps-man configuration> add name=slave-cfg ssid=slave security=wpa2psk security.passphrase=87654321
Create provisioning rule that matches any radio and creates dynamic interfaces using master-cfg and slave-cfg:

[admin@CM] /caps-man provisioning> add action=create-dynamic-enabled
master-configuration=master-cfg
slave-configurations=slave-cfg



data rate
Whether to allow the use of short guard interval (refer to 802.11n MCS specification to see how this may affect throughput). "any" will use either short or long, depending on data rate, "long" will use long only.

Provisioning CAP Interfaces (single and dual band APs)
CAPsMAN distinguishes between CAPs based on a common-name identifier. The identifier is generated based on the following rules:

if CAP provided a certificate, the identifier is set to the Common Name field in the certificate otherwise identifier is based on Base-MAC provided by CAP in the form: '[XX:XX:XX:XX:XX:XX]'.
When the DTLS connection with CAP is successfully established (which means that CAP identifier is known and valid), CAPsMAN makes sure there is no stale connection with CAP using the same identifier. Currently connected CAPs are listed in /caps-man remote-cap menu:

[admin@CM] /caps-man> remote-cap print

```
 # ADDRESS                              IDENT          STATE              RADIOS
 0 00:0C:42:00:C0:32/27044              MT-000C4200C032 Run                1
```

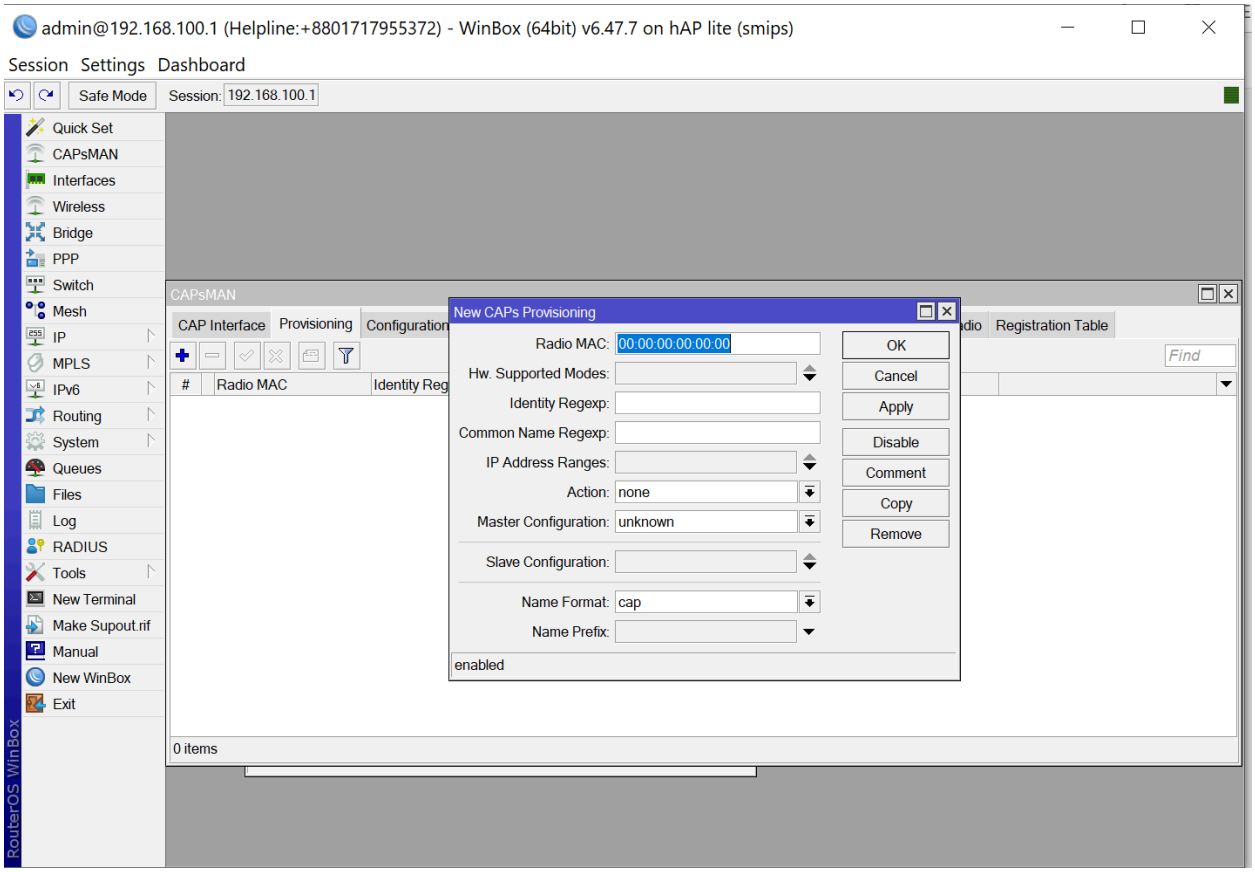
CAPsMAN distinguishes between actual wireless interfaces (radios) based on their builtin MAC address (radio-mac). This implies that it is impossible to manage two radios with the same MAC address on one CAPsMAN. Radios currently managed by CAPsMAN (provided by connected CAPs) are listed in /caps-man radio menu:

```
[admin@CM] /caps-man> radio print
Flags: L - local, P - provisioned
 #    RADIO-MAC         INTERFACE                    REMOTE-AP-IDENT
 0  P 00:03:7F:48:CC:07 cap1                         MT-000C4200C032
```

When CAP connects, CAPsMAN at first tries to bind each CAP radio to CAPsMAN master interface based on radio-mac. If an appropriate interface is found, radio gets set up using master interface configuration and configuration of slave interfaces that refer to particular master interface. At this moment interfaces (both master and slaves) are considered bound to radio and radio is considered provisioned.

If no matching master interface for radio is found, CAPsMAN executes 'provisioning rules'. Provisioning rules is an ordered list of rules that contain settings that specify which radio to match and settings that specify what action to take if a radio matches.

Provisioning rules for matching radios are configured in /caps-man provisioning menu:



Dynamic vs static CAP interfaces on CAPsMAN

Interfaces on CAPsMAN can be static or dynamic. Static interfaces are stored in RouterOS configuration and will persist across reboots. Dynamic interfaces exist only while a particular CAP is connected to CAPsMAN.

Virtual AP (additional SSIDs)

It is possible to create centralized Access Point management setup for home or office environment that is scalable to many Access Point, such a setup is quite easy to configure and has been explained in the Simple CAPSMAN setup guide, but for more complex setups VLANs might be required. CAPsMAN has a functionality to assign a certain VLAN ID under certain conditions. This guide will provide an example how assign a VLAN ID to Wireless packets based on the AP, to which a Wireless client connects to. CAPsMAN with VLANs can be achieved either by using Local Forwarding Mode or CAPsMAN Forwarding Mode, the Local Forwarding Mode will provide the possibility to use a switch between your APs and CAPsMAN router to switch packets (to achieve larger throughput), while CAPsMAN Forwarding Mode should be used when all traffic should always be forwarded to the CAPsMAN router (in most cases to filter packets).

In this example we are going to assign all our Wireless clients to VLAN10, if they connect to WiFi_WORK, and going to assign Wireless clients to VLAN20, if they connect to WiFi_GUEST.

We are going to use Virtual APs along with CAPsMAN to create multiple SSIDs for our Wireless clients to connect to while using a single physical device. An example how to use a single SSID for a single physical device will also be shown by using CAPsMAN provisioning rules.

Using Local Forwarding Mode
Alt text
CAPsMAN topology for VLANs and using Local Forwarding Mode
In Local Forwarding Mode the CAPsMAN router is distributing the configuration across all CAPs that are being provisioned by the CAPsMAN router. In Local Forwarding Mode traffic is not required to be sent to the CAPsMAN router, rather it can be sent to a different router without involving the CAPsMAN router when forwarding traffic. This mode allows you to tag traffic to a certain VLAN ID before it is being sent to your network from your Wireless client, which adds possibilities to use a switch to limit certain VLAN IDs to certain ports. In Local Forwarding Mode traffic is not encapsulated with a special CAPsMAN header, which can only be removed by a CAPsMAN router.

CAPsMAN Router
Create appropriate CAP configurations for each VLAN
```
/caps-man configuration
add country=latvia datapath.local-forwarding=yes datapath.vlan-id=10
datapath.vlan-mode=use-tag name=Config_WORK
security.authentication-types=wpa-psk,wpa2-psk \
   security.passphrase=secret_work_password ssid=WiFi_WORK
add country=latvia datapath.local-forwarding=yes datapath.vlan-id=20
datapath.vlan-mode=use-tag name=Config_GUEST security.authentication-types=\
   wpa-psk,wpa2-psk security.passphrase=secret_guest_password ssid=WiFi_GUEST
```
We are going to create a single CAPsMAN provisioning rule to create the WiFi_WORK and the WiFi_GUEST SSIDs on a single device, each connect CAP is going to create these SSIDs automatically
```
/caps-man provisioning
add action=create-dynamic-enabled master-configuration=Config_WORK
slave-configurations=Config_GUEST
```
Icon-note.png
Note: You can create even more Virtual APs by adding multiple slave-confiugrations. That requires multiple CAPsMAN configurations that were created earlier.

For security reasons, limit the CAPsMAN to a single interface
```
/caps-man manager interface
set [ find default=yes ] forbid=yes
add disabled=no interface=ether1
```
Enable the CAPsMAN manager
```
/caps-man manager
```

set enabled=yes
Setup DHCP Server for each VLAN
/interface vlan
add interface=ether1 name=VLAN10 vlan-id=10
add interface=ether1 name=VLAN20 vlan-id=20
/ip address
add address=192.168.10.1/24 interface=VLAN10
add address=192.168.20.1/24 interface=VLAN20
/ip pool
add name=dhcp_pool10 ranges=192.168.10.2-192.168.10.254
add name=dhcp_pool20 ranges=192.168.20.2-192.168.20.254
/ip dhcp-server
add address-pool=dhcp_pool10 disabled=no interface=VLAN10 name=dhcp10
add address-pool=dhcp_pool20 disabled=no interface=VLAN20 name=dhcp20
/ip dhcp-server network
add address=192.168.10.0/24 dns-server=8.8.8.8 gateway=192.168.10.1
add address=192.168.20.0/24 dns-server=8.8.8.8 gateway=192.168.20.1
Switch
In this example we are going to be using Bridge VLAN Filering to filter unknown VLANs and to assign other devices to the same networks. Some devices are capable of offloading this to the built-in switch chip, check Basic VLAN switching guide to see how to configure it on different types of devices.

Setup Bridge VLAN Filtering
/interface bridge
add name=bridge1 vlan-filtering=yes
/interface bridge port
add bridge=bridge1 interface=ether1
add bridge=bridge1 interface=ether2
add bridge=bridge1 interface=ether3
add bridge=bridge1 interface=ether4 pvid=10
add bridge=bridge1 interface=ether5 pvid=20
/interface bridge vlan
add bridge=bridge1 tagged=ether1,ether2,ether3 untagged=ether4 vlan-ids=10
add bridge=bridge1 tagged=ether1,ether2,ether3 untagged=ether5 vlan-ids=20
Icon-note.png
Note: In this example untagged traffic is going to be used to communicate between CAPs and CAPsMAN Router. By default, if PVID is not changed, untagged traffic is going to be forwarded between ports that have the same PVID value set (including the default PVID).

CAPs
Create a bridge and assign a port to it, that is connect to the CAPsMAN Router

/interface bridge
add name=bridge1
/interface bridge port
add bridge=bridge1 interface=ether1
Enable CAP mode on the AP, make sure you specify to use the newly created bridge
/interface wireless cap
set bridge=bridge1 discovery-interfaces=bridge1 enabled=yes interfaces=wlan1
After CAPs are successfully connected to the CAPsMAN Router, the wlan1 (SSID WiFi_WORK) and a newly created virtual wlan5 (SSID WiFi_GUEST) interfaces get dynamically added as bridge ports. The VLAN is assigned for a wireless interface and as a result all data coming from wireless gets tagged and only data with this tag will be sent out over wireless. A bridge vlan-filtering can be disabled if additional VLAN managing and controlling is not needed. The associated VLAN can be seen with a port VLAN ID (PVID) property.
[admin@CAP_1] /interface bridge port pr
Flags: X - disabled, I - inactive, D - dynamic, H - hw-offload

#		INTERFACE	BRIDGE	HW	PVID	PRIORITY	PATH-COST	INTERNAL-PATH-COST	HORIZON
0	H	ether1	bridge1	yes	1	0x80	10	10	none
1	D	wlan1	bridge1		10	0x80	10	10	none
2	D	wlan5	bridge1		20	0x80	10	10	none

That is it! Connect Wireless clients to your APs and check connectivity.

Using CAPsMAN Forwarding Mode
Alt text
CAPsMAN topology for VLANs and using CAPsMAN Forwarding Mode
In CAPsMAN Forwarding Mode all traffic that is coming from a CAP is encapsulated with a special CAPsMAN header, which can only be removed by a CAPsMAN router, this means that a switch will not be able to distinguish the VLAN ID set by the CAP since the VLAN tag is also going to be encapsulated. This mode limits the possibility to divert traffic in Layer2 networks, but gives you the possibility to forward traffic from each CAP over Layer3 networks for a distant CAPsMAN router to process the traffic, this mode is useful when you want to control multiple CAPs in remote locations, but want to use a central gateway.

CAPsMAN Router
Setup Bridge VLAN filtering to limit interfaces to appropriate VLANs
/interface bridge
add name=bridge1 vlan-filtering=yes
/interface bridge port
add bridge=bridge1 interface=ether1 pvid=10
add bridge=bridge1 interface=ether2 pvid=20
/interface bridge vlan
add bridge=bridge1 tagged=bridge1 untagged=ether1 vlan-ids=10
add bridge=bridge1 tagged=bridge1 untagged=ether2 vlan-ids=20

Icon-note.png
Note: CAPsMAN will attach CAP interfaces to the bridge and automatically will add appropriate entries to the bridge VLAN table. This feature is available starting with RouterOS v6.43

Create appropriate CAP configurations for each VLAN

```
/caps-man configuration
add country=latvia datapath.bridge=bridge1 datapath.vlan-id=10 datapath.vlan-mode=use-tag
name=Config_WORK security.authentication-types=wpa-psk,wpa2-psk \
   security.passphrase=secret_work_password ssid=WiFi_WORK
add country=latvia datapath.bridge=bridge1 datapath.vlan-id=20 datapath.vlan-mode=use-tag
name=Config_GUEST security.authentication-types=wpa-psk,wpa2-psk \
   security.passphrase=secret_guest_password ssid=WiFi_GUEST
```

We are going to create a single CAPsMAN provisioning rule to create the WiFi_WORK and the WiFi_GUEST SSIDs on a single device, each connect CAP is going to create these SSIDs automatically

```
/caps-man provisioning
add action=create-dynamic-enabled master-configuration=Config_WORK
slave-configurations=Config_GUEST
```

Icon-note.png
Note: You can create even more Virtual APs by adding multiple slave-confiugrations. That requires multiple CAPsMAN configurations that were created earlier.

For security reasons, limit the CAPsMAN to interfaces. to which CAPs are going to be connected

```
/caps-man manager interface
set [ find default=yes ] forbid=yes
add disabled=no interface=ether3
add disabled=no interface=ether4
```

Enable the CAPsMAN manager

```
/caps-man manager
set enabled=yes
```

Setup DHCP Server for each VLAN

```
/interface vlan
add interface=bridge1 name=VLAN10 vlan-id=10
add interface=bridge1 name=VLAN20 vlan-id=20
/ip address
add address=192.168.10.1/24 interface=VLAN10
add address=192.168.20.1/24 interface=VLAN20
/ip pool
add name=dhcp_pool10 ranges=192.168.10.2-192.168.10.254
add name=dhcp_pool20 ranges=192.168.20.2-192.168.20.254
```

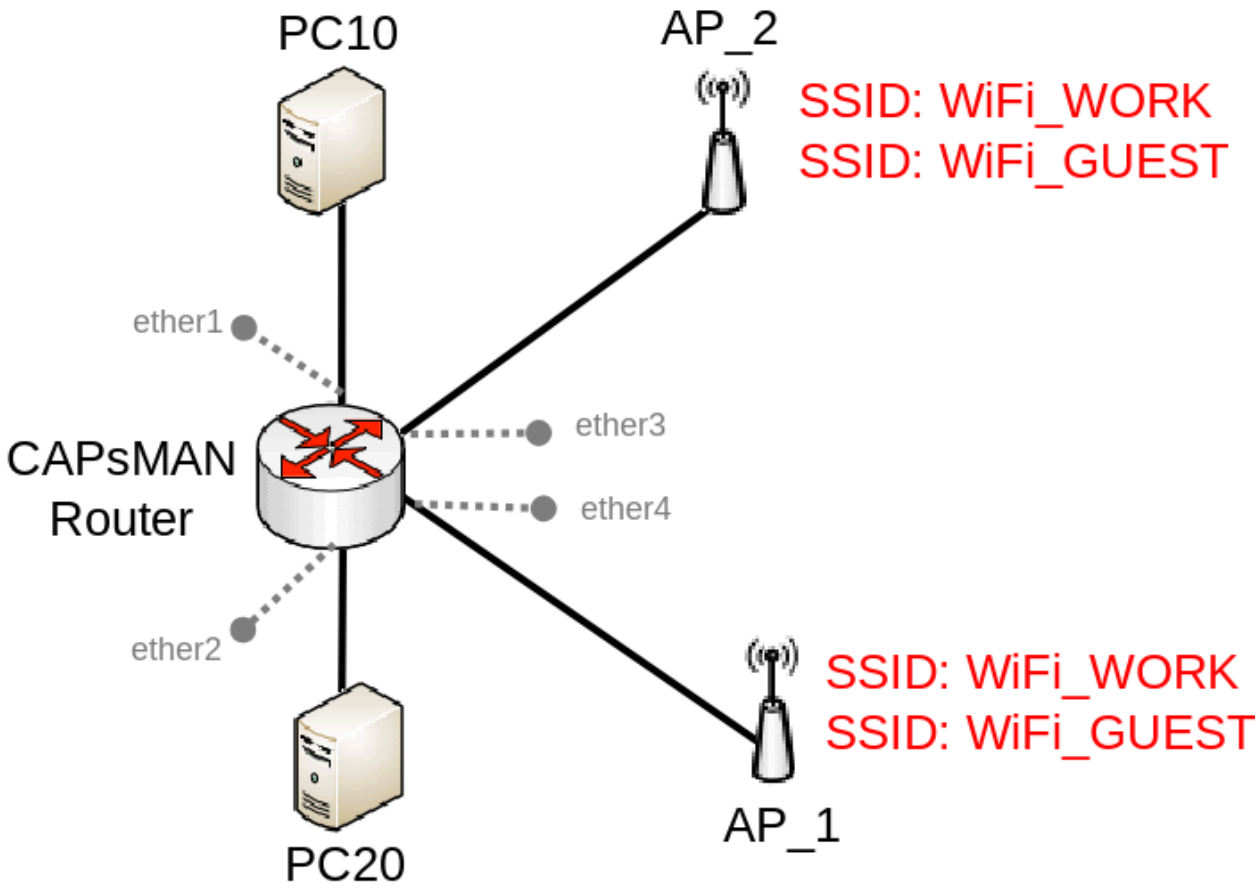
/ip dhcp-server
add address-pool=dhcp_pool10 disabled=no interface=VLAN10 name=dhcp10
add address-pool=dhcp_pool20 disabled=no interface=VLAN20 name=dhcp20
/ip dhcp-server network
add address=192.168.10.0/24 dns-server=8.8.8.8 gateway=192.168.10.1
add address=192.168.20.0/24 dns-server=8.8.8.8 gateway=192.168.20.1
CAPs
Enable CAP mode on each AP, specify which interface is connected to the CAPsMAN router
/interface wireless cap
set discovery-interfaces=ether1 enabled=yes interfaces=wlan1
After CAPs are successfully connected to the CAPsMAN Router, two CAP interfaces will be dynamically created on the CAPsMAN Router. Both of these interfaces will get dynamically added as bridge ports on the same CAPsMAN Router due to explicitly selecting the bridge interface with datapath.bridge=bridge1 and using the default CAPsMAN Forwarding Mode datapath.local-forwarding=no. Because of using a bridge with enabled vlan-filtering, both CAP interfaces will also show up in a bridge VLAN table.
[admin@CAPsMAN_Router] /interface bridge port print
Flags: X - disabled, I - inactive, D - dynamic, H - hw-offload

#	INTERFACE	BRIDGE	HW	PVID	PRIORITY	PATH-COST	INTERNAL-PATH-COST	HORIZON
0	ether1	bridge1	yes	10	0x80	10	10	none
1	ether2	bridge1	yes	20	0x80	10	10	none
2	D cap16	bridge1		10	0x80	10	10	none
3	D cap17	bridge1		20	0x80	10	10	none

[admin@CAPsMAN_Router] /interface bridge vlan print
Flags: X - disabled, D - dynamic

#	BRIDGE	VLAN-IDS	CURRENT-TAGGED	CURRENT-UNTAGGED
0 D	bridge1	1		bridge1
1	bridge1	10	cap16	ether1
2	bridge1	20	cap17	ether2

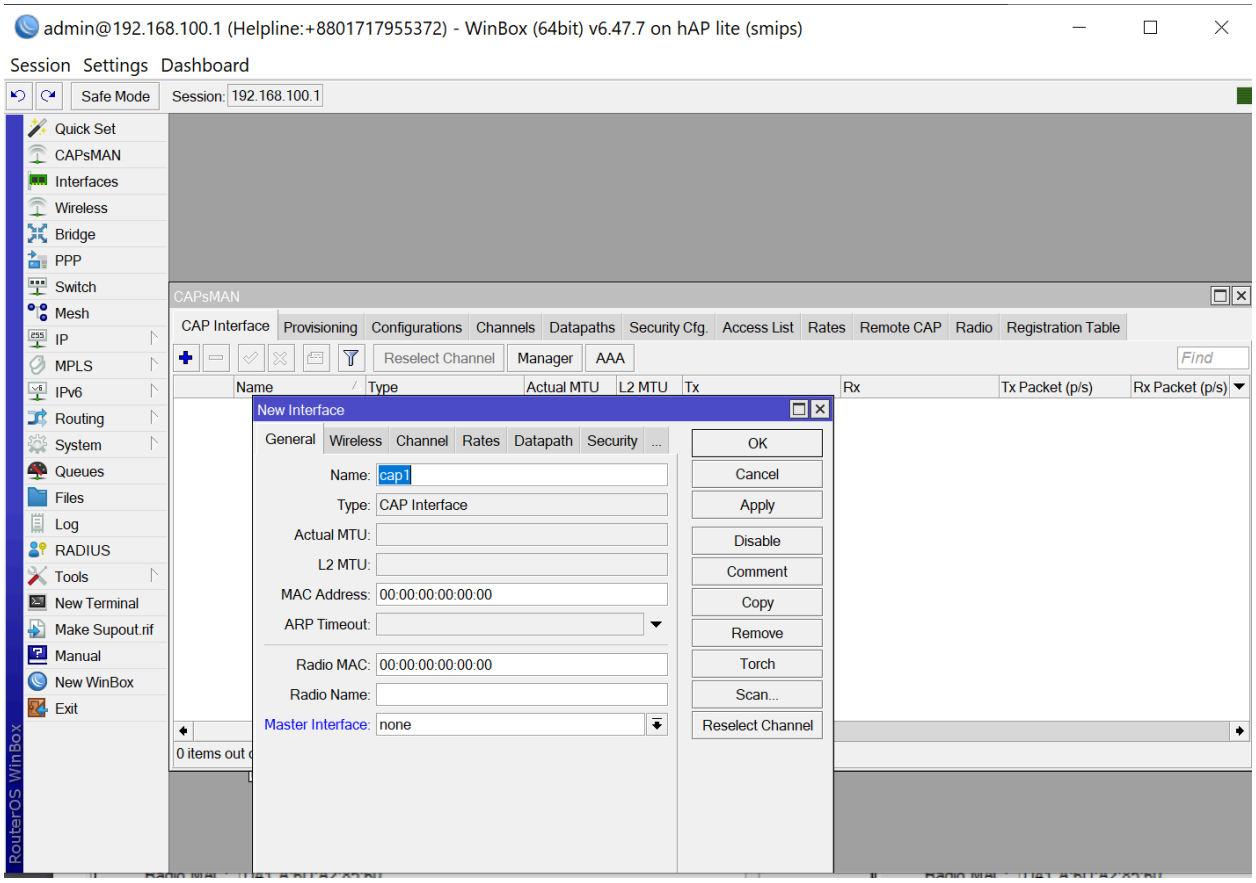
That is it! Connect Wireless clients to your APs and check connectivity.



Static interfaces on CAPs (slave virtual interfaces with VLANs)

No interface name change or setting change after the reboot

- Additional manual setting override
- Copy dynamic interface to make static interface



Access List

Access list on CAPsMAN is an ordered list of rules that is used to allow/deny clients to connect to any CAP under CAPsMAN control. When a client attempts to connect to a CAP that is controlled by CAPsMAN, CAP forwards that request to CAPsMAN. As a part of the registration process, CAPsMAN consults access lists to determine if clients should be allowed to connect. The default behaviour of the access list is to allow connection.

Access list rules are processed one by one until a matching rule is found. Then the action in the matching rule is executed. If action specifies that client should be accepted, client is accepted, potentially overriding it's default connection parameters with ones specified in access list rule.

Access list is configured in /caps-man access-list menu. There are the following parameters for access list rules:

client matching parameters:
address - MAC address of client
mask - MAC address mask to apply when comparing client address
interface - optional interface to compare with interface to which client actually connects to
time - time of day and days when rule matches

signal-range - range in which client signal must fit for rule to match
allow-signal-out-of-range - option which permits client's signal to be out of the range always or for some time interval
action parameter - specifies action to take when client matches:
accept - accept client
reject - reject client
query-radius - query RADIUS server if particular client is allowed to connect
connection parameters:
ap-tx-limit - tx speed limit in direction to client
client-tx-limit - tx speed limit in direction to AP (applies to RouterOS clients only)
client-to-client-forwarding - specifies whether to allow forwarding data received from this client to other clients connected to the same interface
private-passphrase - PSK passphrase to use for this client if some PSK authentication algorithm is used
radius-accounting - specifies if RADIUS traffic accounting should be used if RADIUS authentication gets done for this client
vlan-mode - VLAN tagging mode specifies if traffic coming from client should get tagged (and untagged when going to client).
vlan-id - VLAN ID to use if doing VLAN tagging.
/caps-man access-list

